Calibration Report 08-17-26

Overview

This report analyzes longitudinal calibration data to understand the typical, variable, stationary, and nonstationary behavior of calibration series.

The overall workflow is:

1. Setup: Load and clean the calibration data
2. Exploratory Data Analysis: Extract statistical features and compare similarity using K-means, correlation, and greedy selection
3. Stationarity Testing: Identify stationarity trends and visualize volatile series across geographic region
4. Change Point Analysis: Identify change points on nonstationary series
5. Geographic Representation: Effect of Methodology on Calibration Sample
6. White Noise and Stationary Classification using Ljung-Box and Augumented Dickey-Fuller Tests
7. Applying Moving Average Smoothing for "Ground Truth"
8. Interpretation: Determining a Calibration Threshold
9. Next Steps

1. Setup

1.1 Data Loading

Loads required R packages, imports the geographic landmark metadata, and reads calibration CSV files containing daily calibration rates from all anchors between January 6th, 2026 and April 14th, 2026.

1.2 Cleaning Function

The calibration files contain anchor ID and calibration rate information in slope "m" and intercept "b" values. The cleaning function reallocates the automatically read-in column names as the first row, separates the second column into slope and intercept columns, and combines the two numeric components into a single calibration value by using the $y = mx + b$ equation with an arbitrary $x = 100$. The resulting col4 variable is y, the calibration value used throughout the remainder of the analysis.

1.2.1 Keep IDs Common Across Calibration Files

Not every calibration ID necessarily appears in every file. To choose a sample of anchors that are sufficiently represented across daily measurements, an anchor is retained when its ID appears in at least 80% of the available files. This reduces the likelihood that missing observations dominate time series analysis and across-anchor behavior. 900 of 961 of anchors are retained.

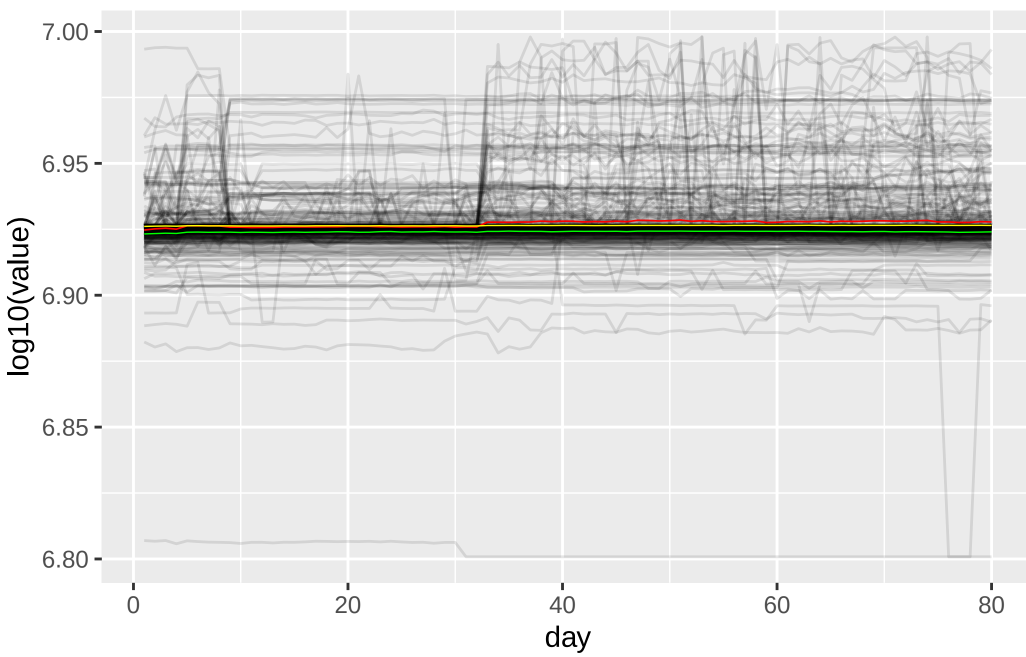
1.2.2 Build the Longitudinal Calibration Dataset

The calibration CSVs are reduced to anchor ID and daily calibration values, with rows corresponding to calibration IDs and columns corresponding to days. There are 98 days of sequential calibration data, but due to days with large volumes of missing data, 18 days were omitted. Namely, a dozen days of calibration data were omitted from the beginning of January, and the remaining days were omitted between January 30th and April 14th, the final day of calibration data collection. Years worth of daily data would be ideal for this study, but we introduce our first calibration rate observations with a three month period. Among the 900 anchor IDs that appeared in at least 80% of our data, only IDs without missing values across all 80 remaining days are retained, leaving 444 of 900 anchor IDs for analysis. The wide calibration dataset is converted to long format so that each row represents one ID-day observation, and so that only one dataframe must be read in for plotting visualizations. Geographic landmark information is then joined to each calibration series–allowing later analyses to examine calibration behavior by continent, country, or ASN.

2. Exploratory Data Analysis

2.1 Overall Calibration Behavior (Spaghetti Plot)

The first visualization shows all calibration trajectories simultaneously. The individual series are shown with low opacity so that the overall population pattern can be seen. The red line represents the mean calibration value at each time point, the green represents the first quartile, and the yellow represents the third quartile.



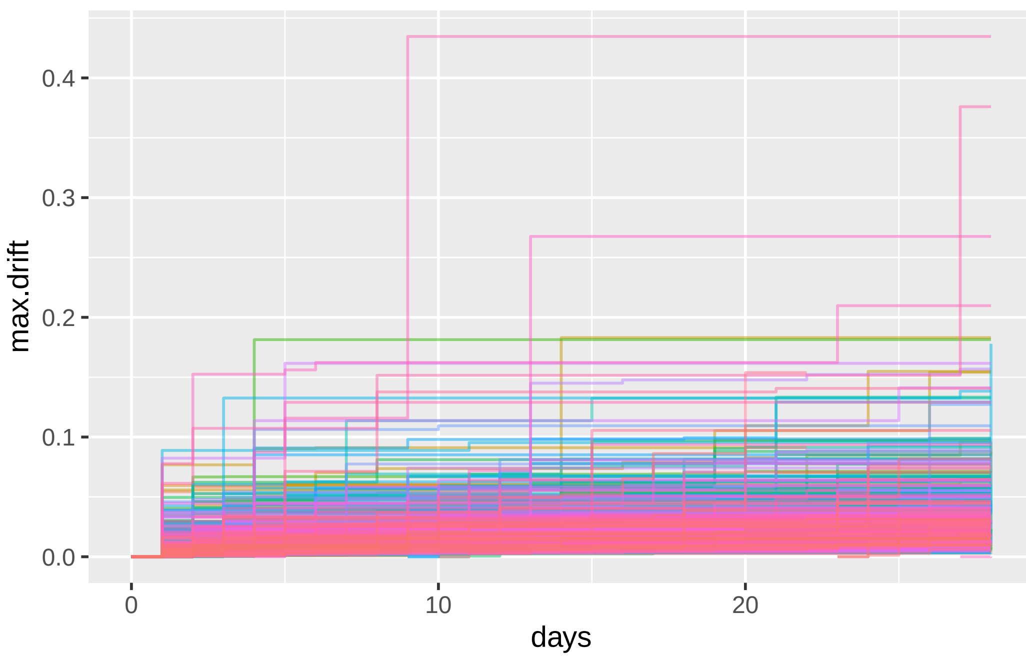
The mean value exceeds the third quartile line once a greater number of anchors demonstrate a sudden jump in calibration rate. The first and third quartile values, and therefore the median value, stay relatively consistent across all 80 days. This demonstrates that while there is a proportion of anchors that deviate from their "ground truth" calibration behavior, that perhaps the ratio of deviating anchors to stable anchors can be assumed to be consistent.

2.1.1 Zach's Calibration Analysis (2024 data)

This aligns with Zach's analysis of the calibration data from March of 2025. In some visualizations Zach sent over to me and Katie over discord, Zach visualizes instantaneous drift from calibration on day 0 over a period of 29 days. The data is from Katie's calibration retrieval between October 21st and November 18th, 2024. It is replicated below:

```
# A tibble: 19,731 x 3
   date       id    calibration
   <date>     <chr>       <dbl>
 1 2024-10-21 7236       73452.
 2 2024-10-21 6232       80942.
 3 2024-10-21 6841       69560.
 4 2024-10-21 6660       72808.
 5 2024-10-21 6368       68017.
 6 2024-10-21 6835       76021.
 7 2024-10-21 7171       58421.
 8 2024-10-21 6396       70384.
 9 2024-10-21 6577       70212.
10 2024-10-21 7020       74485.
# i 19,721 more rows
```

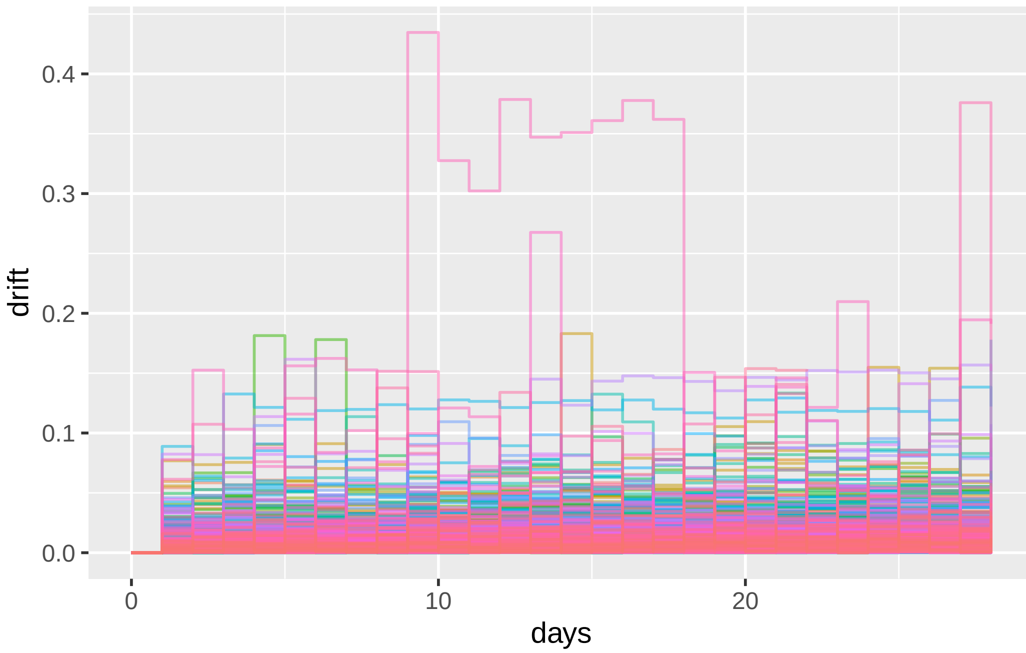
2.1.2 Zach's Plots & Analysis



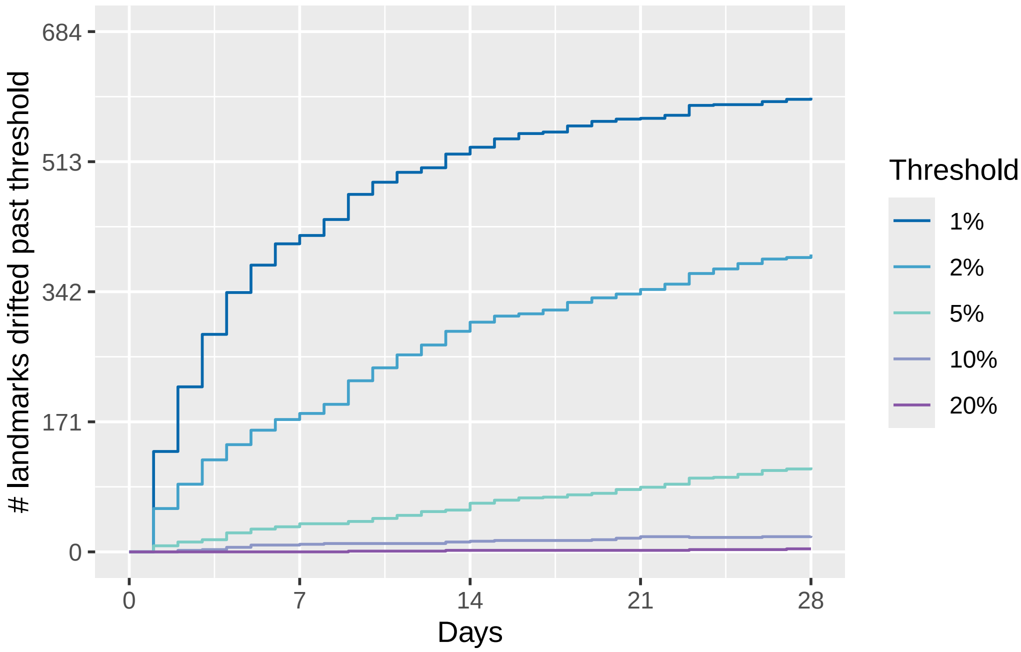
"This is a plot of the maximum deviation of each node's individual calibration from its calibration on day 0, as a function of time. The colors don't mean anything they're just because there's a lot of lines on top of each other here."

"And that's a little hard to understand so here's another way of looking at it. This is the number of nodes that have drifted at least x% away from their calibration on day 0, where x =

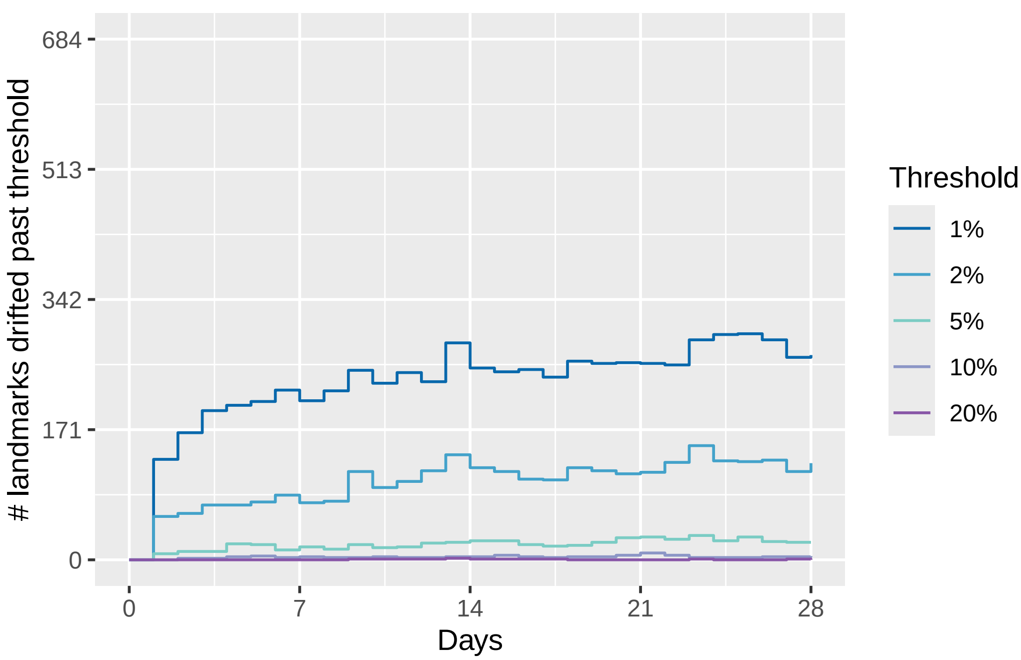
1, 2, 5, 10, 20 as indicated in the legend. By the time we get to day 28, most of the nodes are at least 1% off, but the vast majority are still within 5%, and only a tiny handful are off by more than 10%."



"But wait! Nodes might not stay way off. Here's the instantaneous drift from the calibration on day 0, rather than the maximum. You can see that even the few nodes that go way off don't stay there. This looks much more like a steady state process than one that's changing over time."



“Same”other way of looking at it” as before: instantaneous number of nodes that are off by more than X% from day 0.”



“At any given time, the number of nodes who are off by at least X% is … roughly constant.

But which nodes those are changes from day to day. I have to think about it some more but what this is suggesting to me is that instead of redoing the calibration after some fixed number of days we should be thinking about incrementally refining the calibration on a per-node basis over time. Sliding window medians or something. And maybe we could borrow an idea from NTP and tune the frequency of updates for each node based on how much variance we observe for that node."

2.1.3 Adding on to Zach's Analysis

The spaghetti plot from 2.1 that visualizes January 16 to April 14 2026 data and the behaviors of the quartiles demonstrate a similar behavior to Zach's analysis of October 21 to November 18 2024 calibration data. While the 2026 data points span 80 days in comparison to the 29 days of 2024 data, the same conclusions can be made about calibration deviation behavior.

2.2 Feature Extraction

Each calibration series is summarized using several characteristics: mean level, standard deviation, linear trend, minimum, maximum, and lag-1 autocorrelation. These features provide a compact description of the different behaviors in the calibration population and form the basis for the K-means clustering analysis.

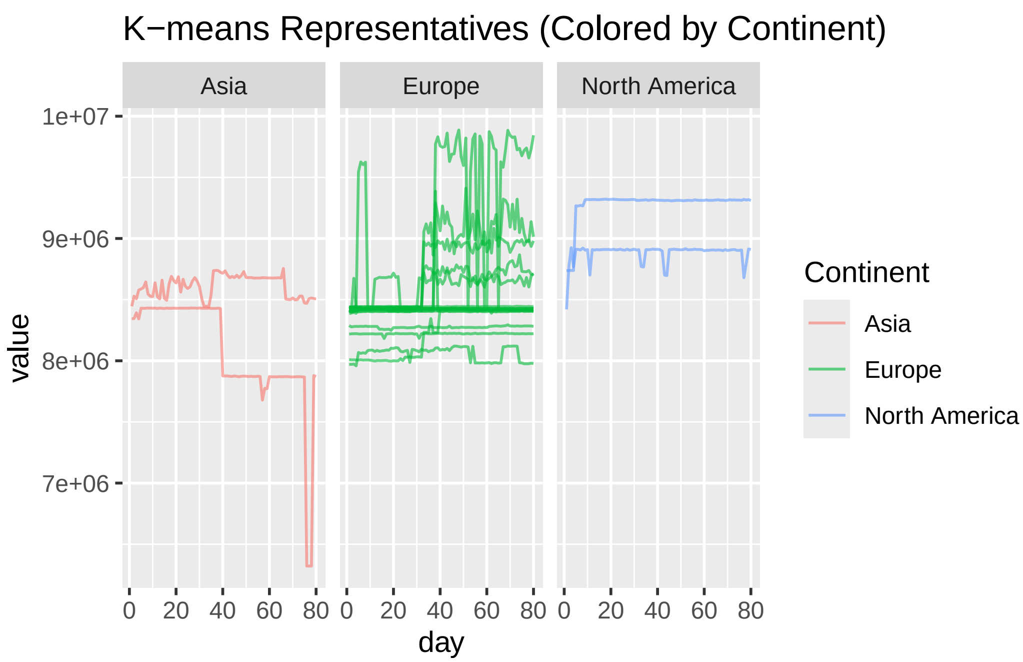
2.3 K-Means Method

2.3.1 K-Means Representative Series

K-means clustering is used to divide the calibration series into 20 groups based on their statistical features. Within each cluster, the series closest to the cluster center is selected as the representative. This provides a smaller set of calibration series that captures the major behavioral patterns in the full population.

2.3.2 Visualize K-Means Representatives

The selected representative series are plotted and colored by continent. This allows the selected behavioral patterns to be examined alongside their geographic distribution.



2.3.3 K-Means Cluster Sizes

For each selected representative, the number of series belonging to its cluster is calculated. This indicates how many calibration series are represented by each selected example.

	ID	cluster	similar_series_count
1	6540	1	47
2	7507	2	46
3	7295	3	4
4	7169	4	7
5	7442	5	13
6	7574	6	7
7	7109	7	42
8	7591	8	10
9	7141	9	16
10	7198	10	2
11	6408	11	7
12	6937	12	52
13	7025	13	65
14	6979	14	9
15	7499	15	9
16	7010	16	17

```
17  6980        17                        51
18  6313        18                         3
19  6352        19                        24
20  6841        20                        13
```

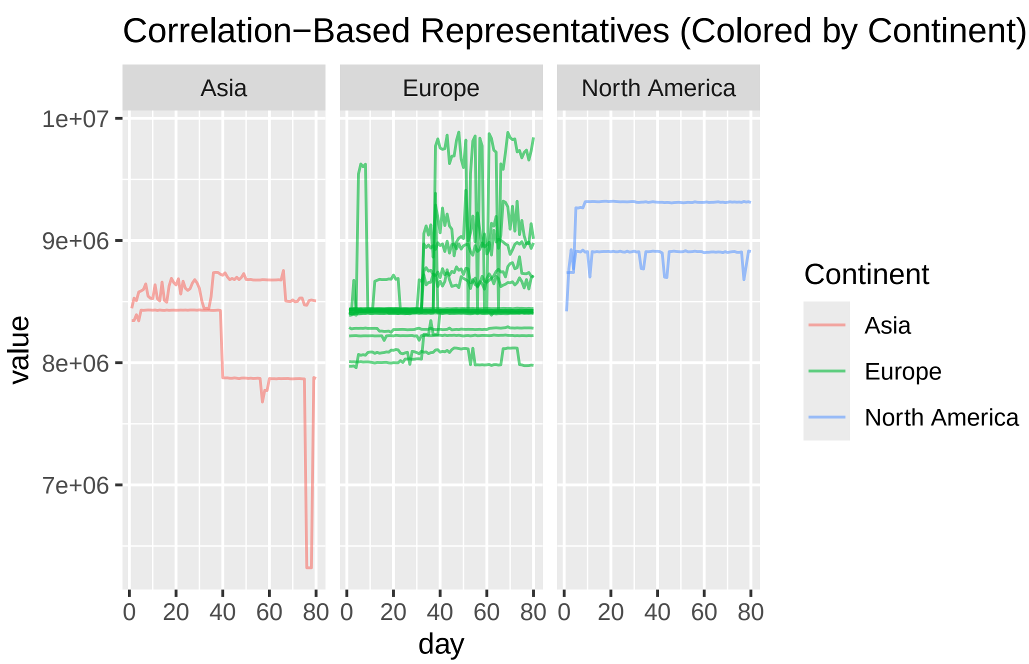
2.4 Correlation Method

2.4.1 Correlation-Based Similarity

A second approach compares calibration series directly using their time-series correlation. A correlation of at least 0.90 is treated as indicating strong similarity. For each K-means representative, the number of highly correlated series is then counted.

2.4.2 Visualize Correlation-Based Representatives

The same K-means representative IDs are visualized using their correlation relationships and geographic metadata.



2.4.3 Correlation Summary

The following table records how many other calibration series are at least 90% correlated with each representative.

	ID	similar_series_count
6540	6540	1
7507	7507	0
7295	7295	20
7169	7169	18
7442	7442	0
7574	7574	0
7109	7109	0
7591	7591	1
7141	7141	0
7198	7198	0
6408	6408	1
6937	6937	3
7025	7025	0
6979	6979	21
7499	7499	3
7010	7010	0
6980	6980	0
6313	6313	1
6352	6352	0
6841	6841	0

2.5 Greedy Method

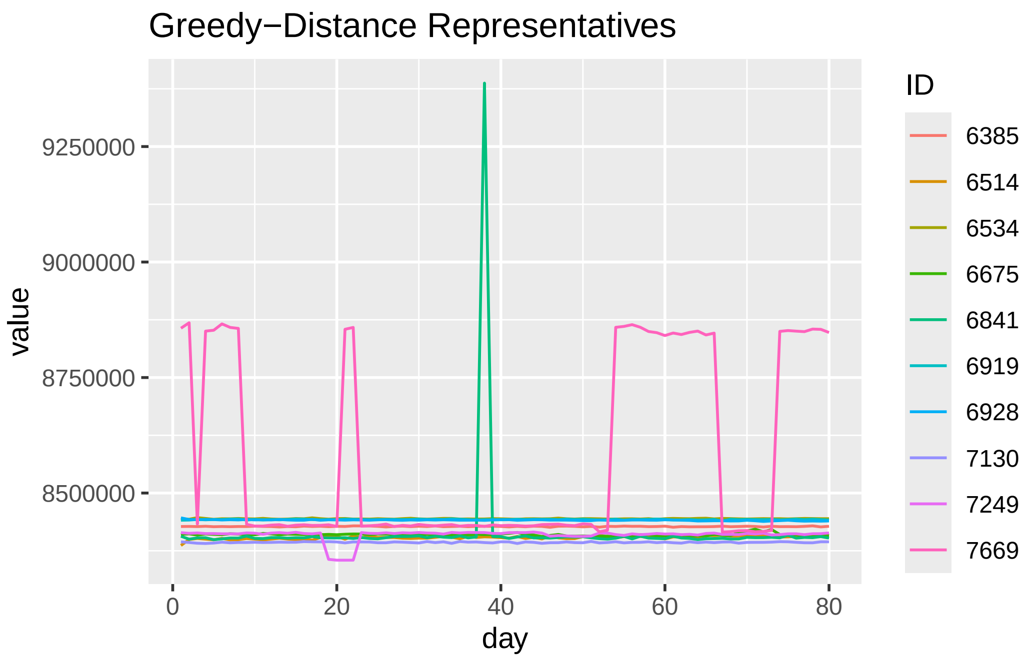
2.5.1 Greedy Representative Selection

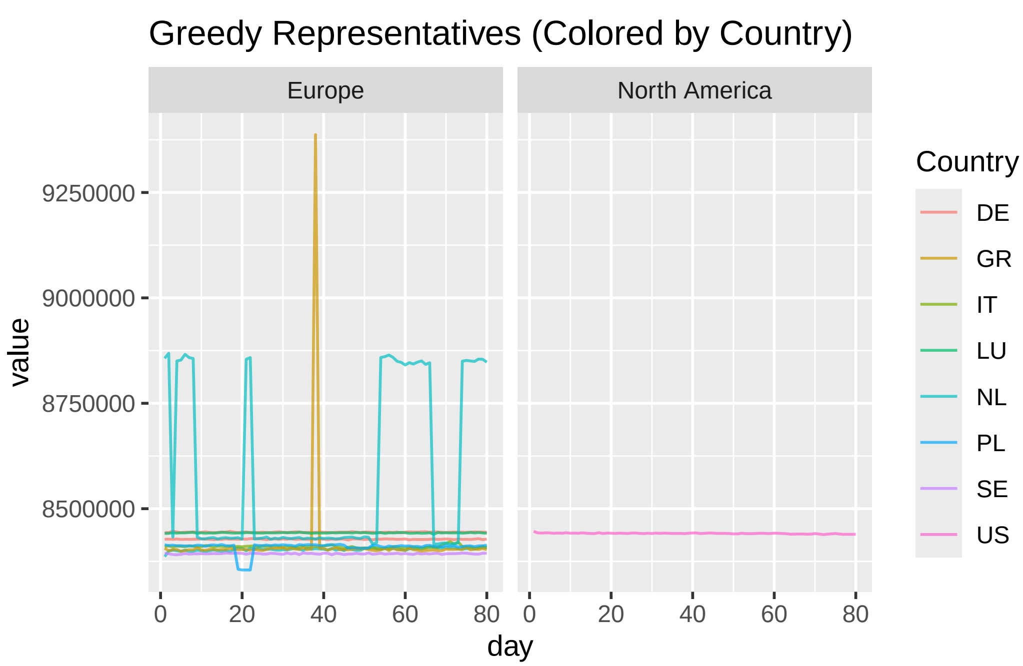
A third approach selects a small number of series that are maximally different from one another.

Correlation is converted into a distance measure using: distance = 1 - correlation. The greedy procedure first selects the most globally separated series and then iteratively chooses the series that is farthest from the already-selected representatives.

2.5.2 Visualize Greedy Representatives

The selected series are plotted individually and then with geographic information to examine the diversity captured by the greedy method.





2.5.3 Similarity Counts for Greedy Representatives

A threshold based on the 20th percentile of the distance matrix is used to count the number of relatively similar series associated with each greedy representative.

```
       ID similar_series_count
6928 6928                   77
6514 6514                  173
6919 6919                   32
7249 7249                   39
7669 7669                   71
7130 7130                   17
6675 6675                   25
6534 6534                   17
6841 6841                    8
6385 6385                   13
```

2.6 Geographic Information for Representative Series

Geographic metadata are added to the K-means, correlation, and greedy representative summaries so that the selected series can be interpreted in their geographic context.

```
     ID similar_series_count        city country    asn     continent
1  6928                   77      Reston      US  57695 North America
2  6514                  173   Rotterdam      NL  24586        Europe
3  6919                   32 Bettembourg      LU  49624        Europe
4  7249                   39   Bialystok      PL  41931        Europe
5  7669                   71     Dronten      NL 213615        Europe
6  7130                   17     Grillby      SE 210422        Europe
7  6675                   25    Piacenza      IT 201333        Europe
8  6534                   17   Frankfurt      DE 200462        Europe
9  6841                    8      Athens      GR 199399        Europe
10 6385                   13    Mannheim      DE  21473        Europe
```

2.7 Build Similarity Tables

For each representative, the analysis creates a table of the individual IDs that are considered similar to that representative. Separate tables are produced for the greedy, K-means, and correlation-based methods.

2.7.1 Greedy Based

	ID	city	country	asn	continent	representative
1	6307	Amsterdam	NL	3333	Europe	6928
2	6317	Zürich	CH	559	Europe	6928
3	6329	Milano	IT	42473	Europe	6928
4	6330	Helsinki	FI	203296	Europe	6928
5	6348	Amsterdam	NL	3333	Europe	6928
6	6372	Neubiberg	DE	680	Europe	6928

2.7.2 K Means Based

	ID	city	country	asn	continent	representative
1	6251	Tampere	FI	719	Europe	6540
2	6329	Milano	IT	42473	Europe	6540
3	6330	Helsinki	FI	203296	Europe	6540
4	6450	Rome	IT	35574	Europe	6540
5	6482	Fremont	US	7247	North America	6540
6	6514	Rotterdam	NL	24586	Europe	6540

2.7.3 Correlation-Based

	ID	city	country	asn	continent	representative
1	6523	Münchenstein	CH	206291	Europe	6540
2	6540	Basel	CH	206484	Europe	6540
3	7507	Strasbourg	FR	61136	Europe	7507
4	6325	Vienna	AT	48943	Europe	7295
5	6360	Salzburg	AT	39878	Europe	7295
6	6435	Rozzano	IT	35574	Europe	7295

2.8 Geographic Composition of K-Means Clusters

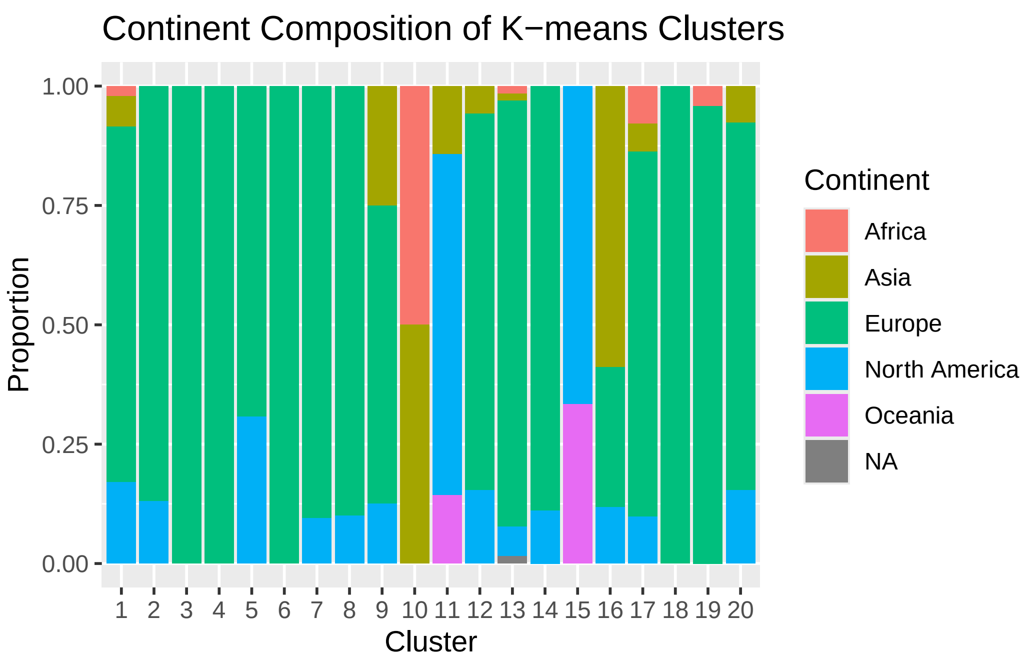
The next analysis asks whether behavioral clusters correspond to geographic regions. For every K-means cluster, the proportion of observations belonging to each continent is calculated. Cluster purity is defined as the proportion of the cluster belonging to its most common continent. A purity of 1 indicates that a cluster consists entirely of one continent, while lower values indicate more geographically mixed clusters.

```
# A tibble: 20 x 4
   cluster purity dominant_continent cluster_size
```

```
     <int>  <dbl> <chr>                     <int>
 1      10  0.5   Africa                        2
 2      16  0.588 Asia                         17
 3       9  0.625 Europe                       16
 4      15  0.667 North America                 9
 5       5  0.692 Europe                       13
 6      11  0.714 North America                 7
 7       1  0.745 Europe                       47
 8      17  0.765 Europe                       51
 9      20  0.769 Europe                       13
10      12  0.788 Europe                       52
11       2  0.870 Europe                       46
12      14  0.889 Europe                        9
13      13  0.892 Europe                       65
14       8  0.9   Europe                       10
15       7  0.905 Europe                       42
16      19  0.958 Europe                       24
17       3  1     Europe                        4
18       4  1     Europe                        7
19       6  1     Europe                        7
20      18  1     Europe                        3
```

2.9 Visualize Cluster Geography

The composition of each K-means cluster is visualized as a stacked bar chart. This provides a direct view of how geographically mixed each cluster is.


Continent Composition of K−means Clusters
Proportion
1.00
0.75
0.50
0.25
0.00
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Cluster
Continent
Africa
Asia
Europe
North America
Oceania
NA

3. Stationarity Testing

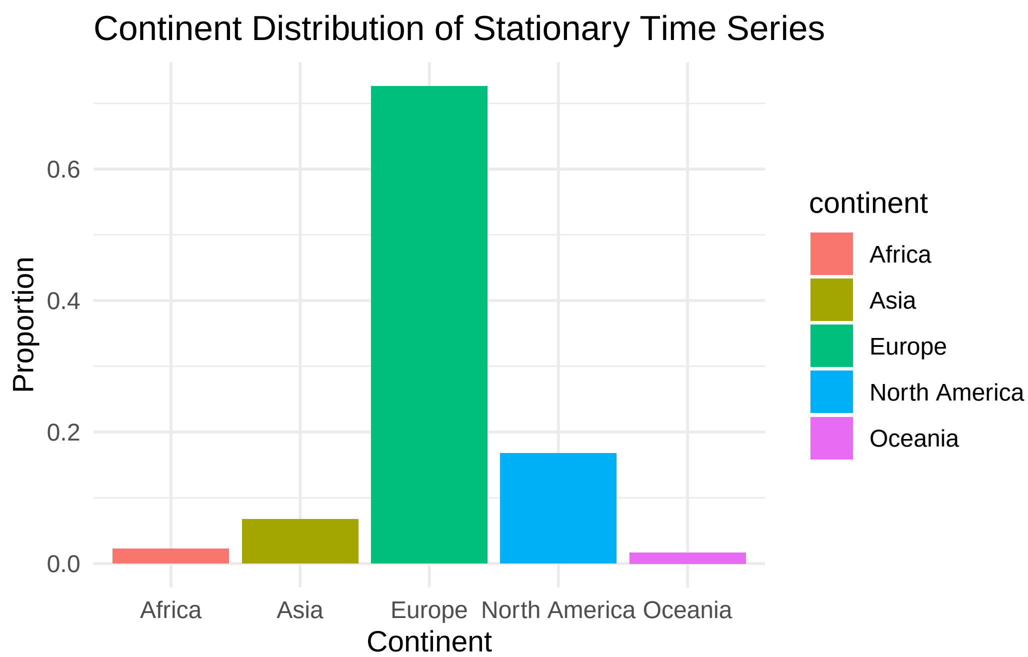
The Augmented Dickey-Fuller (ADF) test is applied independently to every calibration series. A p-value below 0.05 is classified as stationary. This allows the analysis to separate calibration series with relatively stable statistical behavior from series whose underlying behavior changes over time.

3.1 Geographic Distribution of Stationarity

The next analyses examine whether stationary and nonstationary calibration series are distributed differently across continents.

```
# A tibble: 5 x 3
  continent         n proportion
  <chr>         <int>      <dbl>
1 Europe          130     0.726
2 North America    30     0.168
3 Asia             12     0.0670
4 Africa            4     0.0223
5 Oceania           3     0.0168
```

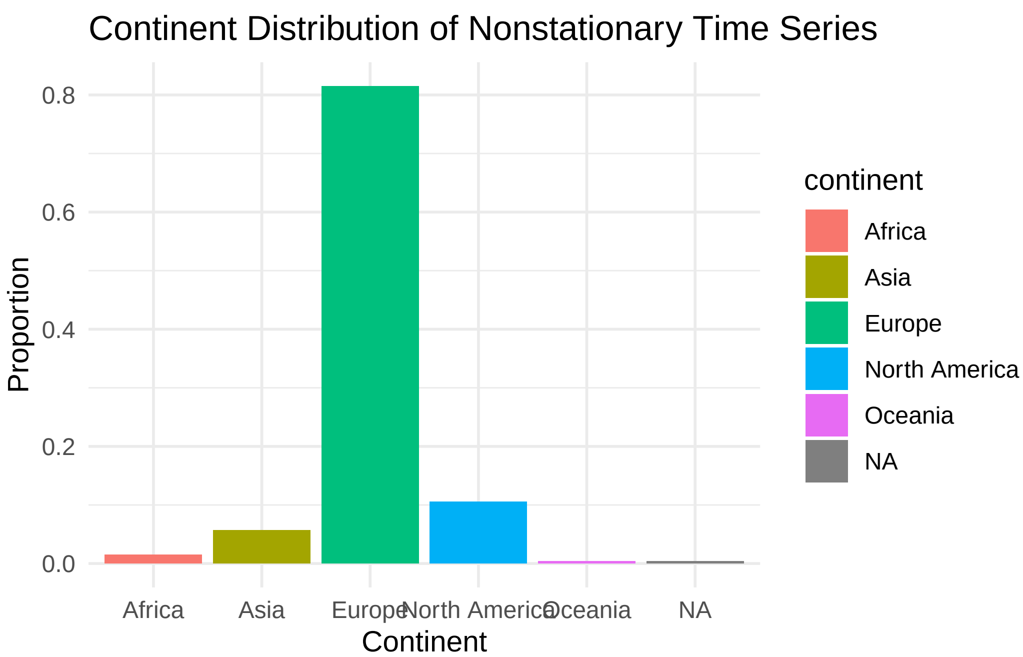
```
# A tibble: 6 x 4
  continent     total stationary proportion_stationary
  <chr>         <int>      <int>                 <dbl>
1 Oceania           4          3                 0.75
2 North America    58         30                 0.517
3 Africa            8          4                 0.5
4 Asia             27         12                 0.444
5 Europe          346        130                 0.376
6 <NA>              1          0                 0
```



3.2 Nonstationary Series

The complementary population of nonstationary series is identified and its geographic composition is examined

```
# A tibble: 6 x 3
  continent         n proportion
  <chr>         <int>      <dbl>
1 Europe          216    0.815
2 North America    28    0.106
3 Asia             15    0.0566
4 Africa            4    0.0151
5 Oceania           1    0.00377
6 <NA>              1    0.00377
```



3.3 Stationarity by Continent

A combined table summarizes the total number of series and the proportion that are stationary or nonstationary within each continent.

```
# A tibble: 6 x 6
  continent     total stationary nonstationary prop_stationary prop_nonstationary
  <chr>         <int>      <int>         <int>           <dbl>              <dbl>
1 <NA>              1          0             1           0                      1
2 Africa            8          4             0           0.5                    0
3 Asia             27         12             0           0.444                  0
4 Europe          346        130             0           0.376                  0
5 North Ameri~     58         30             0           0.517                  0
6 Oceania           4          3             0           0.75                   0
```

```
# A tibble: 6 x 4
  continent      total nonstationary proportion_nonstationary
  <chr>          <int>         <int>                    <dbl>
1 <NA>               1             1                    1
2 Europe           346           216                    0.624
3 Asia              27            15                    0.556
4 Africa             8             4                    0.5
```

```
5 North America    58            28                     0.483
6 Oceania           4             1                     0.25
```

3.4 Volatility of Nonstationary Series

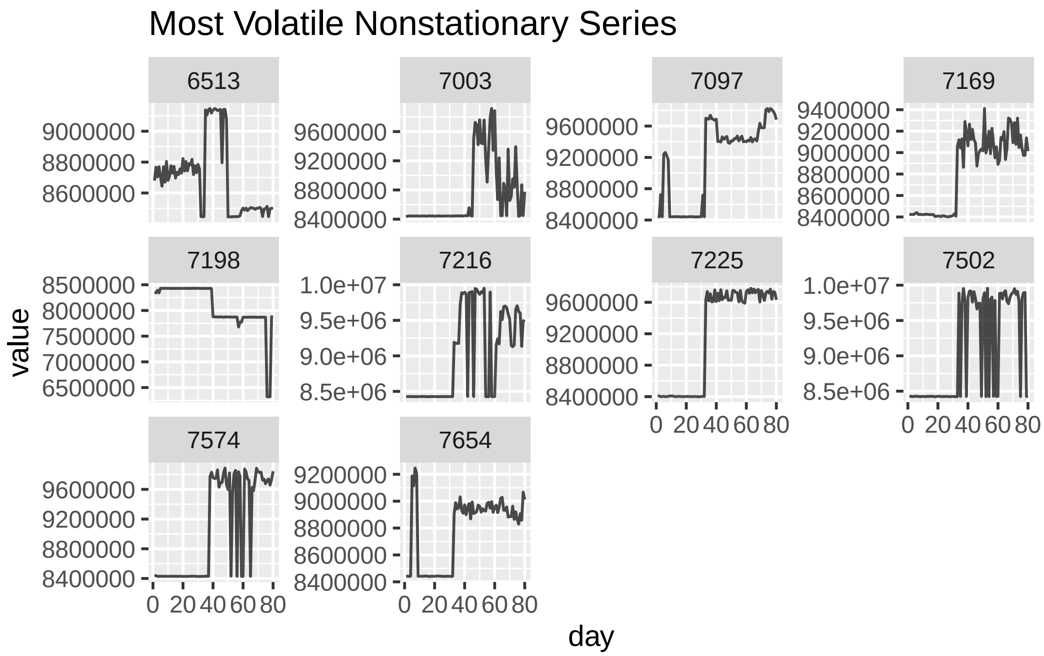
The next step identifies the most highly volatile nonstationary calibration series. Volatility is measured as the mean absolute change between consecutive observations. Standard deviation is also calculated as a second measure of overall variation.

```
# A tibble: 10 x 3
   ID    volatility sd_value
   <chr>      <dbl>    <dbl>
 1 7502     350247.  711551.
 2 7216     201837.  626072.
 3 7574     186606.  668880.
 4 7003     178064.  457248.
 5 7169      96953.  350764.
 6 7097      75633.  528142.
 7 6513      61096.  237349.
 8 7225      60025.  636658.
 9 7654      59364.  257737.
10 7198      54960.  446335.
```

```
# A tibble: 10 x 7
   ID    volatility sd_value city              country   asn continent
   <chr>      <dbl>    <dbl> <chr>             <chr>   <int> <chr>
 1 7502     350247.  711551. Vienna            AT       1764 Europe
 2 7216     201837.  626072. Vienna            AT      35492 Europe
 3 7574     186606.  668880. Vienna            AT      51184 Europe
 4 7003     178064.  457248. Arnhem            NL      39636 Europe
 5 7169      96953.  350764. Frankfurt         DE      39063 Europe
 6 7097      75633.  528142. Frankfurt am Main DE      56381 Europe
 7 6513      61096.  237349. Nanterre          FR      62000 Europe
 8 7225      60025.  636658. Vienna            AT      29287 Europe
 9 7654      59364.  257737. Frankfurt         DE      20292 Europe
10 7198      54960.  446335. Dubai             AE       3491 Asia
```

3.5 Visualize Highly Volatile Series

The ten most volatile nonstationary series are plotted individually so that their behavior can be inspected directly.



3.6 Volatility and Geography

Volatility statistics are joined to geographic metadata to determine how the nonstationary population is distributed across continents.

```
# A tibble: 6 x 2
  continent     n_series
  <chr>            <int>
1 Europe             216
2 North America       28
3 Asia                15
4 Africa               4
5 Oceania              1
6 <NA>                 1
```

```
# A tibble: 6 x 3
  continent     n_series proportion
  <chr>            <int>      <dbl>
1 Europe             216    0.815
2 North America       28    0.106
3 Asia                15    0.0566
4 Africa               4    0.0151
```

```
5 Oceania              1    0.00377
6 <NA>                 1    0.00377
```

3.7 High-Volatility Series by Continent

A series is classified as highly volatile when its volatility exceeds the 90th percentile of volatility across the nonstationary population.

The resulting table shows the number and proportion of high-volatility series within each continent.

```
# A tibble: 6 x 4
  continent     high_vol_count total proportion_high_vol
  <chr>                  <int> <int>               <dbl>
1 Africa                     0     4              0
2 Asia                       3    15              0.2
3 Europe                    22   216              0.102
4 North America              2    28              0.0714
5 Oceania                    0     1              0
6 <NA>                       0     1              0
```

3.8 Stationarity Summary by Continent

This provides another consolidated view of stationary versus nonstationary behavior within each continent.

```
# A tibble: 6 x 6
  continent    total stationary nonstationary prop_stationary prop_nonstationary
  <chr>        <int>      <int>         <int>           <dbl>              <dbl>
1 <NA>             1          0             1           0                      1
2 Africa           8          4             0           0.5                    0
3 Asia            27         12             0           0.444                  0
4 Europe         346        130             0           0.376                  0
5 North Ameri~    58         30             0           0.517                  0
6 Oceania          4          3             0           0.75                   0
```

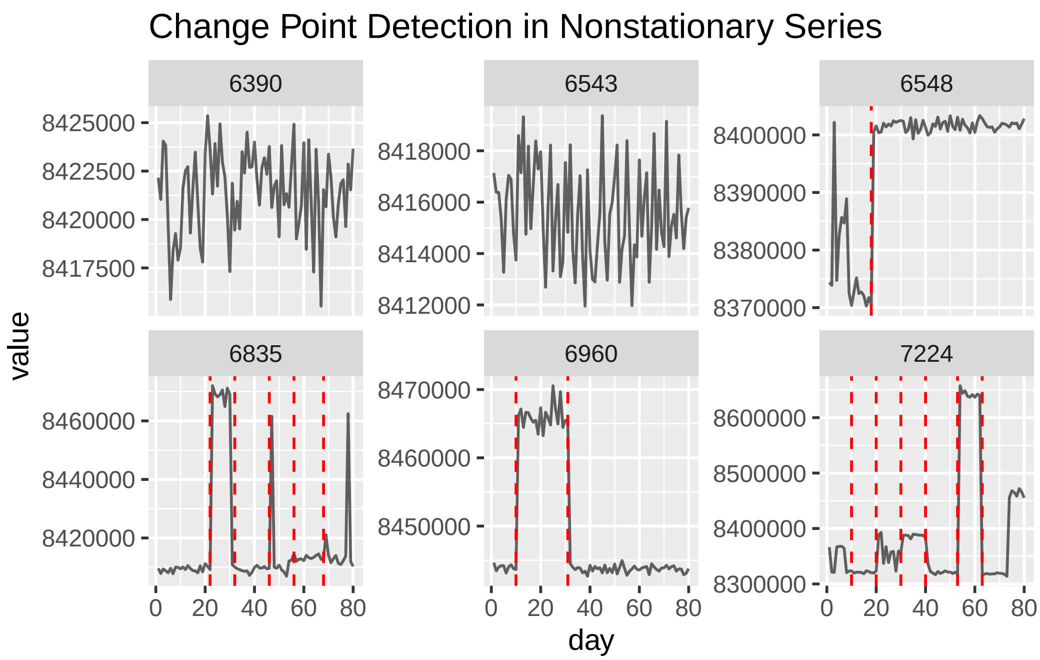
4. Change-Point Analysis

The nonstationary series are examined for discrete changes in their mean or variance.

The PELT method with a BIC penalty is used to identify change points. A minimum segment length of 10 observations is required, and series shorter than 20 observations are skipped.

4.1 Visualize Detected Change Points

Six nonstationary series are sampled for visualization. The detected change points are overlaid on the corresponding time series.




```
# A tibble: 247 x 2
  ID    n_changes
  <chr>     <int>
1 6251          3
2 6307          2
3 6313          4
4 6317          2
5 6323          5
6 6325          2
7 6329          1
8 6330          1
```

```
 9 6331          1
10 6332          3
# i 237 more rows
```

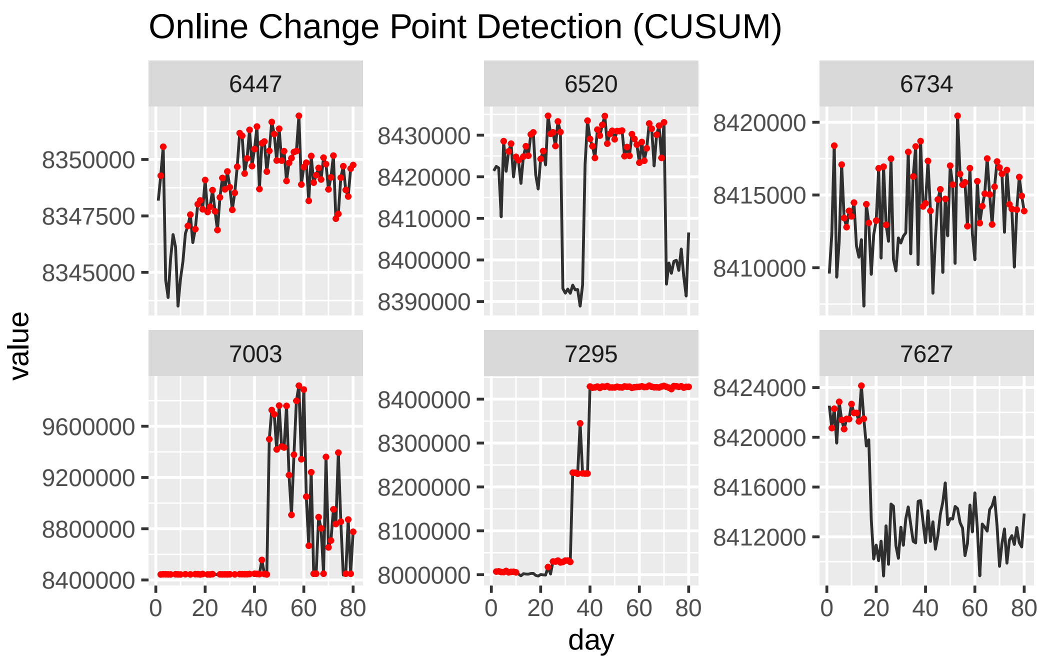
4.2 Online Change-Point Analysis

A second change-detection approach uses a CUSUM-style online procedure.

The first 20 observations establish a baseline mean. Subsequent observations are accumulated into a CUSUM statistic, with an alarm generated when the statistic exceeds the specified threshold. The statistic is reset after an alarm.

4.3 Visualize Online Change Points

The online alarms are plotted for six randomly selected nonstationary series.



5. Geographic Representation Before and After Filtering

The next comparison tracks how many calibration anchors are present in each continent before filtering and how many remain after the QMD's filtering and complete-case requirements.

The final columns then show the proportion of stationary and nonstationary series among the retained observations.

```
# A tibble: 6 x 5
  continent     total_observations selected_observations prop_stationary
  <chr>                      <int>                 <int>           <dbl>
1 Africa                        30                     8           0.5
2 Asia                         159                    27           0.444
3 Europe                       509                   346           0.376
4 North America                180                    58           0.517
5 Oceania                       30                     4           0.75
6 South America                 53                    NA          NA
# i 1 more variable: prop_nonstationary <dbl>
```

6. White Noise and Stationarity Classification

The next stage distinguishes random-looking calibration series from series with detectable temporal structure. The Ljung-Box test is used to identify white-noise behavior, while the ADF test is used to classify stationarity.

The two tests are combined into four initial categories: White noise & stationary, White noise & nonstationary, Structured & stationary, and Structured & nonstationary

```
# A tibble: 4 x 3
  category                          n   prop
  <chr>                         <int>  <dbl>
1 Nonstationary (structured)      225 0.507
2 Stationary (structured)          77 0.173
3 White noise (nonstationary?)     40 0.0901
4 White noise (stationary)        102 0.230
```

6.1 White-Noise Classification

The Ljung-Box test is also isolated so that white-noise series can be analyzed independently of the stationarity classification.

6.2 Overall White-Noise Proportion

The overall proportion of calibration series classified as white noise is calculated.

```
# A tibble: 1 x 3
  total white_noise proportion_white_noise
  <int>       <int>                  <dbl>
1   444         142                  0.320
```

6.3 White Noise by Continent

The proportion of white-noise series is compared across continents.

```
# A tibble: 6 x 4
  continent     total white_noise proportion_white_noise
  <chr>         <int>       <int>                  <dbl>
1 Oceania           4           2                  0.5
2 Africa            8           3                  0.375
```

```
3 North America    58          21                    0.362
4 Europe          346         111                    0.321
5 Asia             27           5                    0.185
6 <NA>              1           0                    0
```

```
# A tibble: 6 x 6
  continent    total stationary nonstationary prop_stationary prop_nonstationary
  <chr>        <int>      <int>         <int>           <dbl>              <dbl>
1 <NA>             1          0             1           0                      1
2 Africa           8          4             0           0.5                    0
3 Asia            27         12             0           0.444                  0
4 Europe         346        130             0           0.376                  0
5 North Ameri~    58         30             0           0.517                  0
6 Oceania          4          3             0           0.75                   0
```

6.4 Stationarity Among Non-White-Noise Series

For series that are not classified as white noise, the ADF test is applied again to explicitly identify which of those series are stationary.

This produces the final three-way classification: White noise (unstructured), Stationary (structured), and Nonstationary (structured).

6.5 Final Classification Tables

The stationary white-noise series and structured stationary series are identified separately, followed by an overall classification summary.

```
# A tibble: 0 x 5
# i 5 variables: ID <chr>, is_white_noise <lgl>, adf_p <dbl>,
#   is_stationary <lgl>, category <chr>
```

```
# A tibble: 77 x 5
   ID    is_white_noise  adf_p is_stationary category
   <chr> <lgl>           <dbl> <lgl>         <chr>
 1 6316  FALSE          0.01   TRUE          Stationary (structured)
 2 6327  FALSE          0.01   TRUE          Stationary (structured)
 3 6328  FALSE          0.01   TRUE          Stationary (structured)
 4 6336  FALSE          0.0490 TRUE          Stationary (structured)
 5 6374  FALSE          0.0431 TRUE          Stationary (structured)
 6 6379  FALSE          0.01   TRUE          Stationary (structured)
```

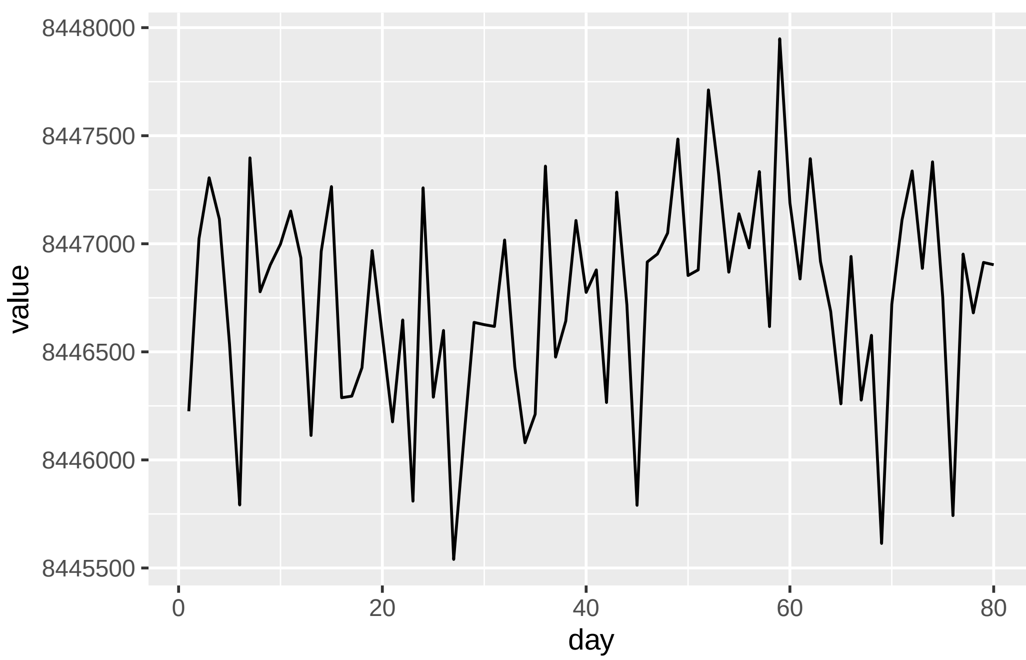
```
 7 6426  FALSE          0.01   TRUE          Stationary (structured)
 8 6446  FALSE          0.0248 TRUE          Stationary (structured)
 9 6450  FALSE          0.0152 TRUE          Stationary (structured)
10 6506  FALSE          0.0156 TRUE          Stationary (structured)
# i 67 more rows

# A tibble: 3 x 3
  category                          n proportion
  <chr>                         <int>      <dbl>
1 Nonstationary (structured)      225      0.507
2 Stationary (structured)          77      0.173
3 White noise (unstructured)      142      0.320
```

6.6 Investigating White-Noise but Nonstationary Series

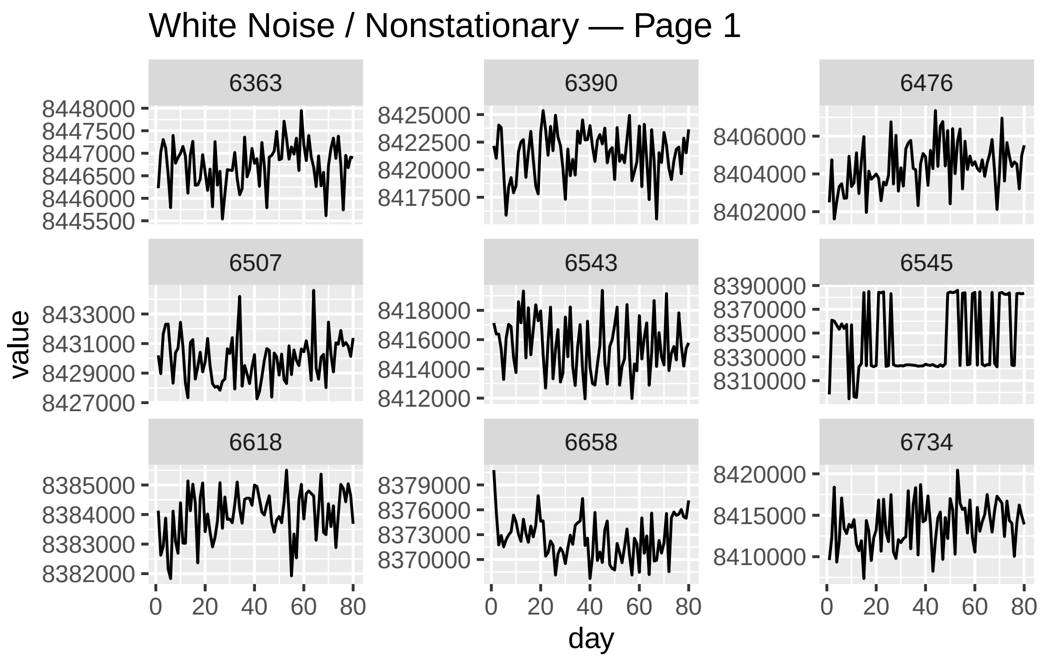
Some series are classified as white noise by the Ljung-Box test but nonstationary by the ADF test.

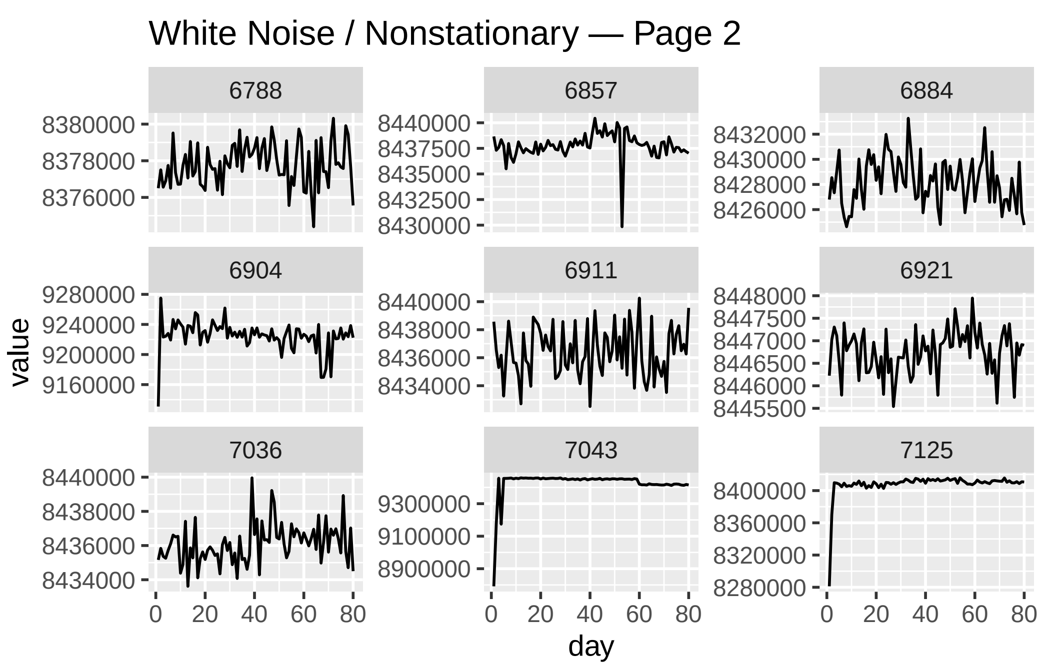
These series are examined visually because the combination of the two tests suggests that they may contain slowly changing behavior that is not easily detected as conventional temporal structure.

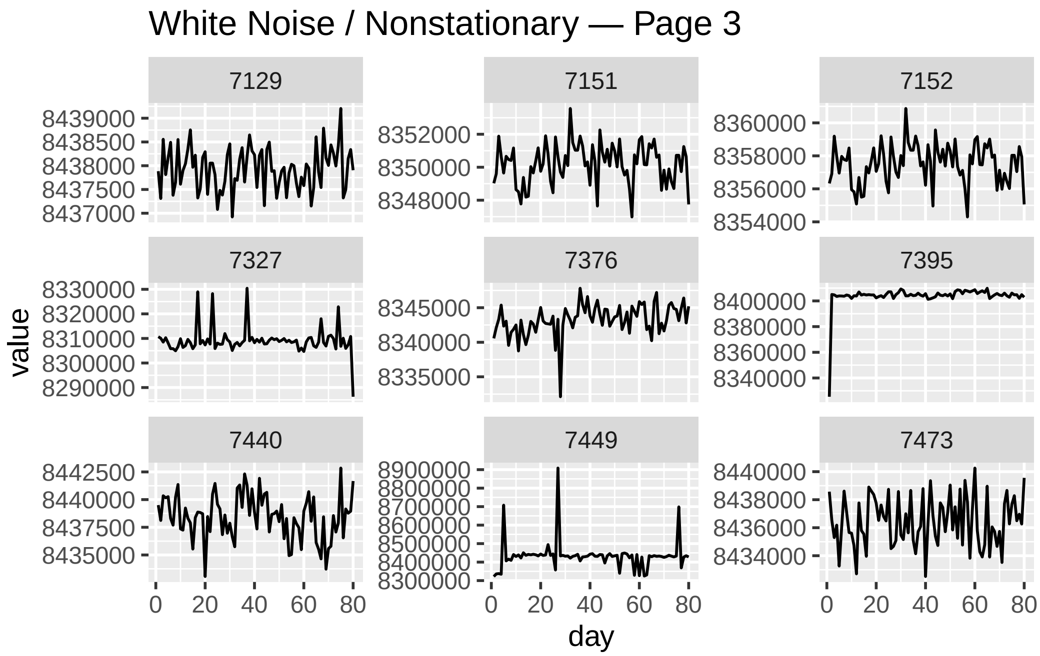


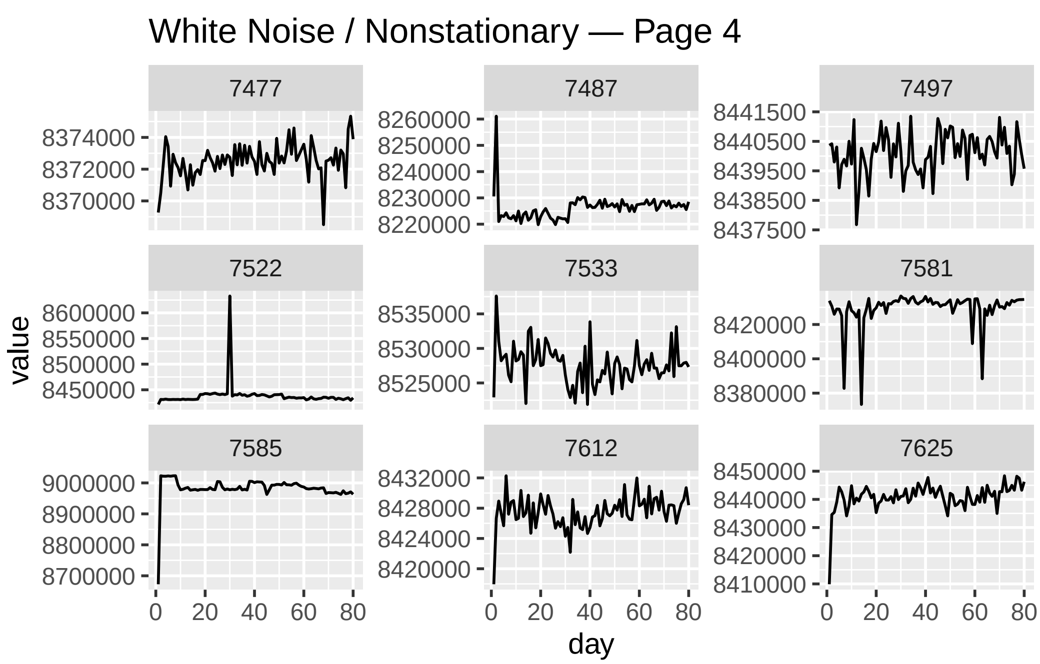
6.7 Visualize All White-Noise / Nonstationary Series

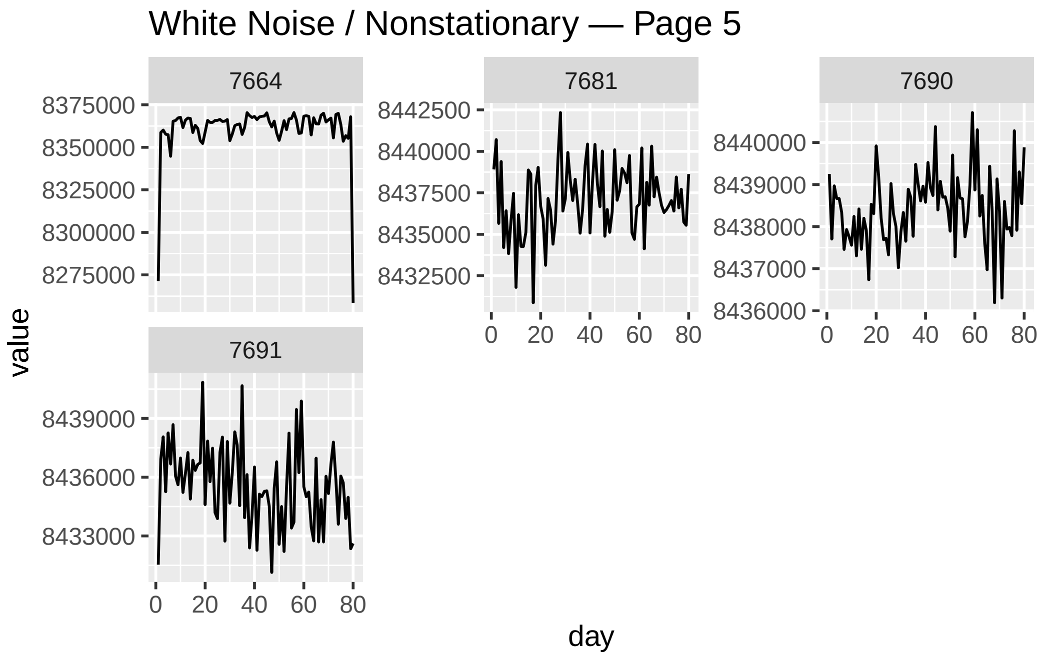
All white-noise/nonstationary series are plotted in groups of nine to allow visual inspection of potential drift, steps, variance changes, or other instability.






White Noise / Nonstationary — Page 3
7129
7151
7152
7327
7376
7395
7440
7449
7473
value
day


White Noise / Nonstationary — Page 4
7477
7487
7497
7522
7533
7581
7585
7612
7625
value
day



6.7.1 Save White-Noise / Nonstationary Visualizations

The same visualizations are saved as PNG files for later inspection.

7. Moving-Average

7.1 Moving-Average Stationarity Check

The white-noise/nonstationary series are smoothed using a centered MA(5). The purpose is to reduce short-term fluctuations and test whether the underlying smoothed signal behaves more like a stationary process.

7.2 Compare Original and Smoothed Stationarity

The original ADF classification is compared with the ADF result after MA(5) smoothing. This determines how many white-noise/nonstationary series become stationary after short-term fluctuations are smoothed out.

```
# A tibble: 40 x 4
   ID     adf_p ma_adf_p ma_stationary
   <chr>  <dbl>    <dbl> <lgl>
 1 6363  0.186    0.227  FALSE
 2 6390  0.0673   0.01   TRUE
 3 6476  0.0553   0.317  FALSE
 4 6507  0.0577   0.0113 TRUE
 5 6543  0.108    0.341  FALSE
 6 6545  0.0854   0.0690 FALSE
 7 6618  0.0552   0.0266 TRUE
 8 6658  0.655    0.383  FALSE
 9 6734  0.0993   0.01   TRUE
10 6788  0.130    0.176  FALSE
# i 30 more rows
```

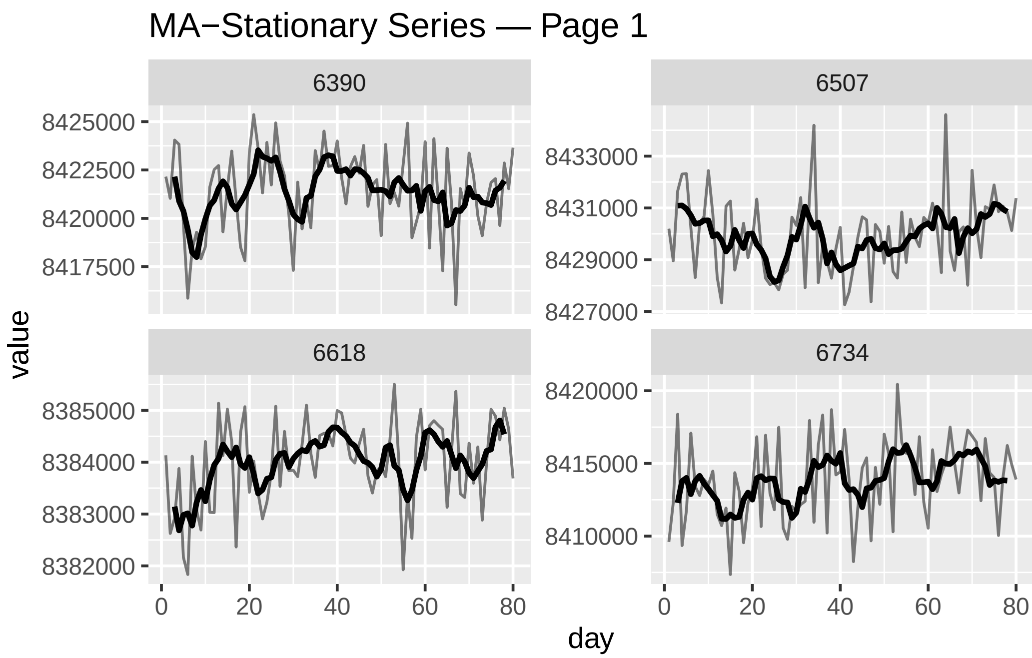
7.3 First Smoothing Stage: Summary of MA(5) Results

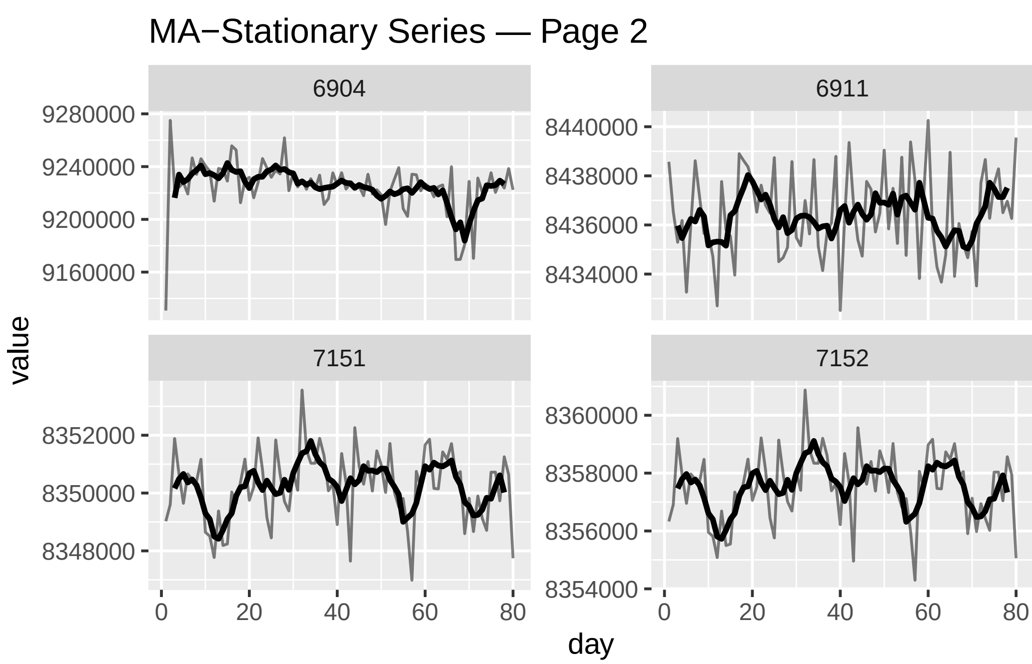
The number and proportion of initially nonstationary white-noise series that become stationary after MA(5) smoothing are calculated.

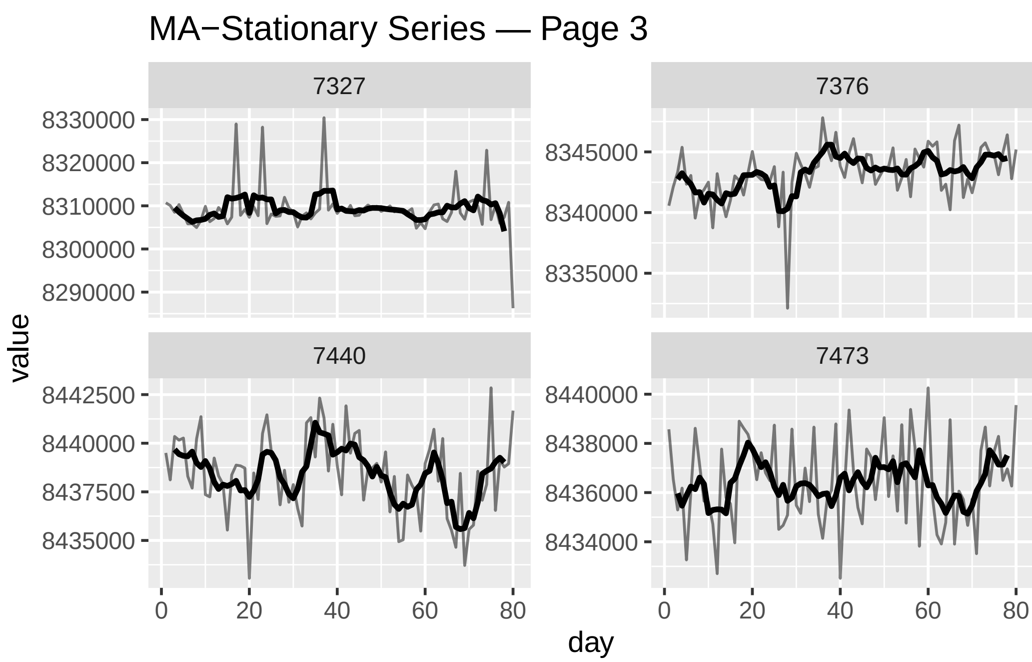
```
# A tibble: 1 x 3
  total become_stationary proportion
  <int>             <int>      <dbl>
1    40                16        0.4
```

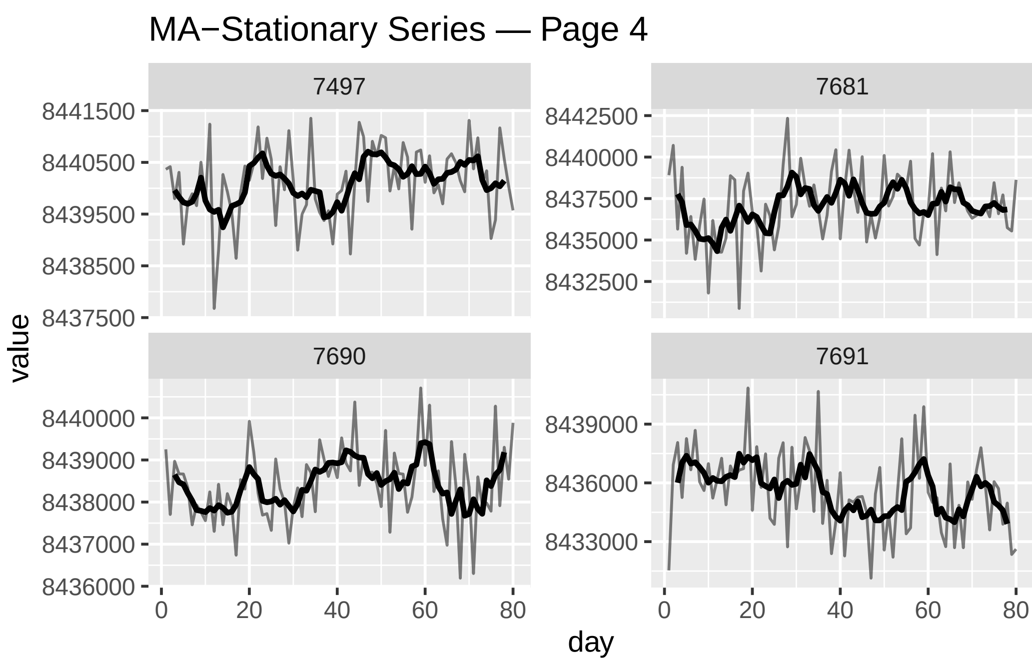
7.3.1 Visualize Series Becoming Stationary After MA(5)

Series that become stationary after MA(5) are visualized with both the original and smoothed signals.









7.3.2 Save MA(5) Visualizations

The MA(5) plots are saved as PNG files.

7.3.3 Remaining White-Noise / Nonstationary Series

The series that remain nonstationary after the first MA(5) smoothing are isolated for additional smoothing.

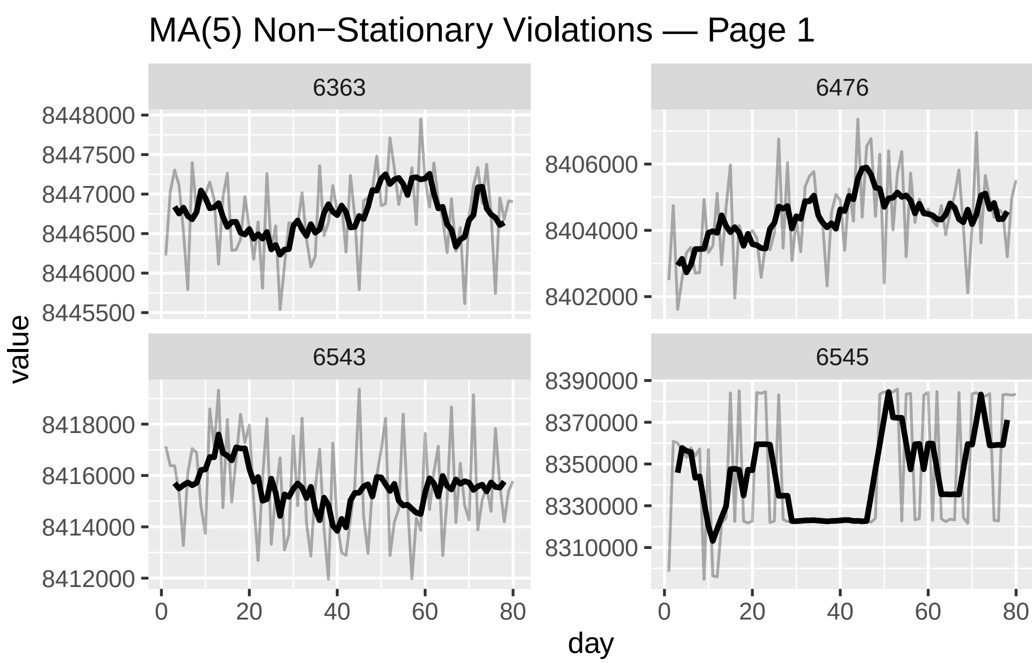
```
[1] 24
```

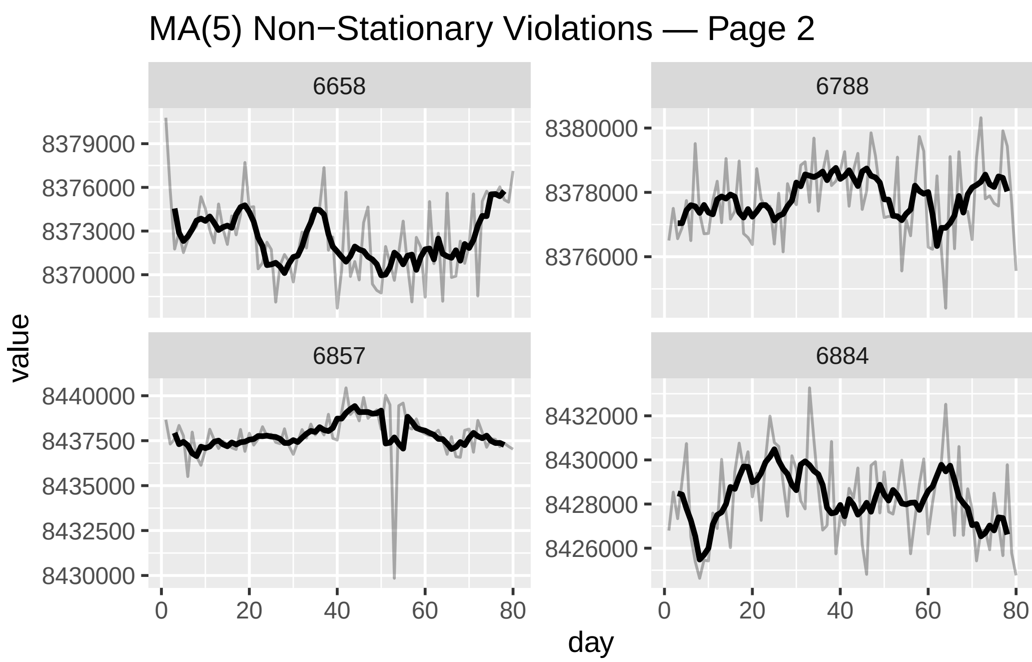
7.4 Second Smoothing Stage: Summary of MA(5) → MA(10) Results

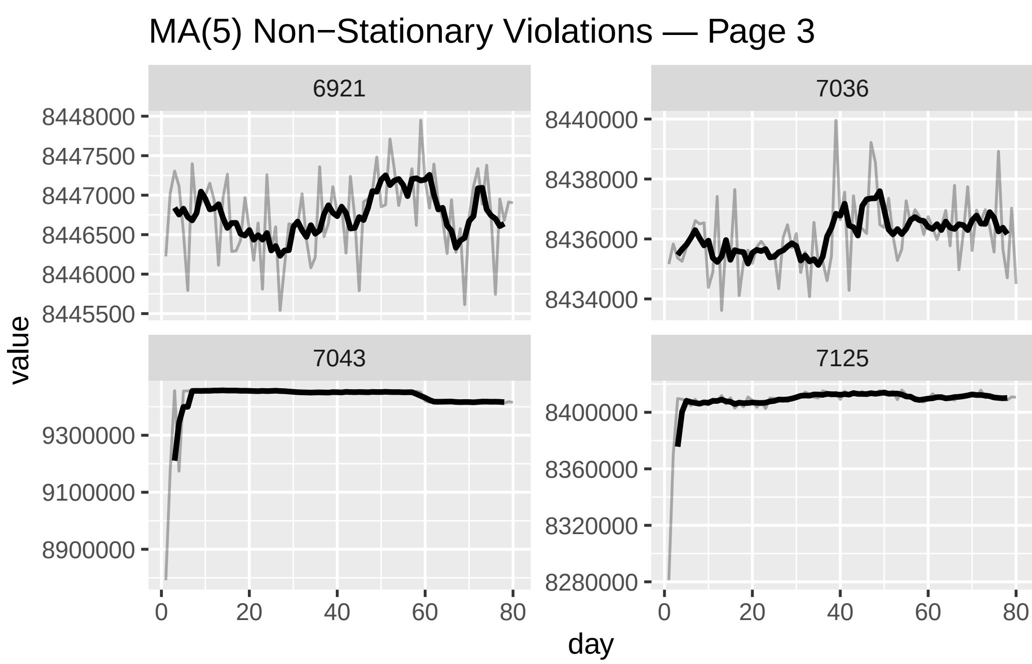
The remaining series are subjected to a second smoothing stage. The first MA(5) reduces short-term variation, and an MA(10) is then applied to the resulting signal. The resulting smoothed series is tested for stationarity again.

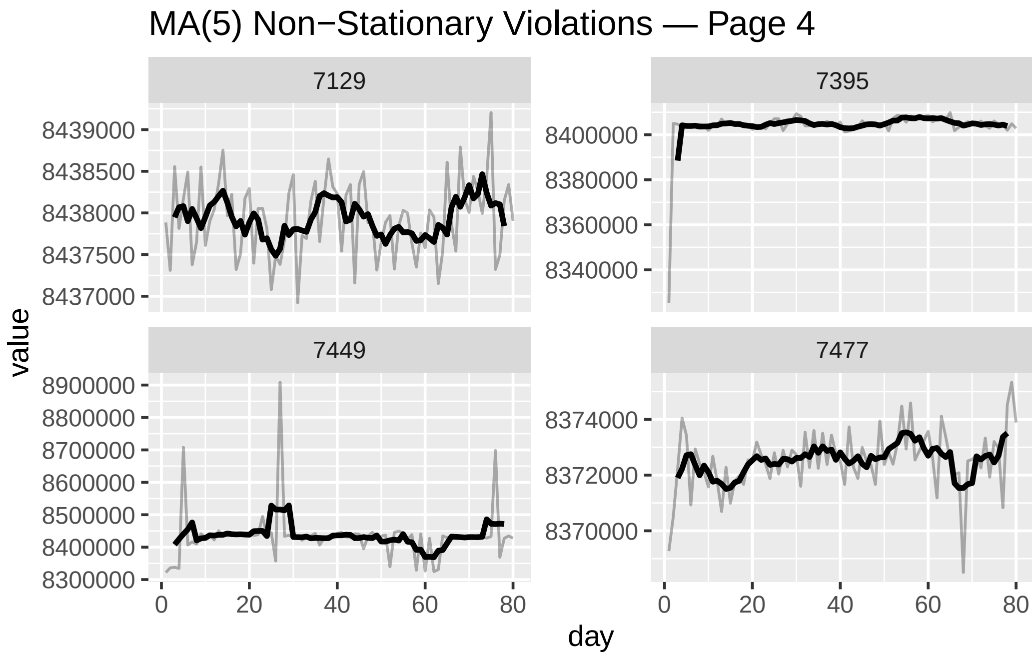
7.4.1 Visualize Remaining MA(5) Violations

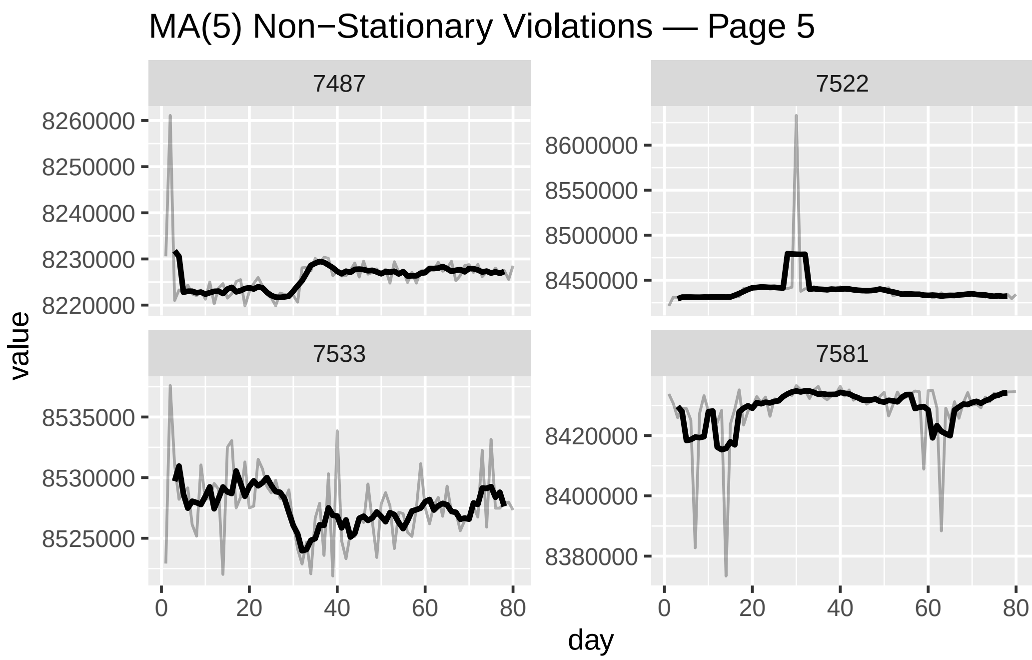
The series that remain nonstationary after the MA(5) stage are plotted with both the original signal and its smoothed version.

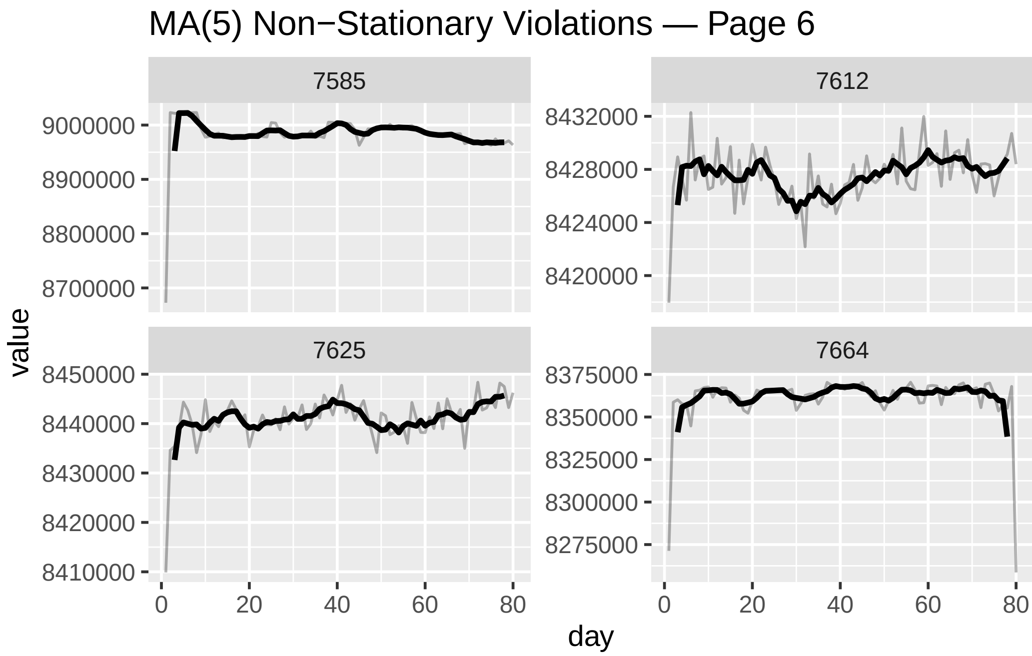



MA(5) Non−Stationary Violations — Page 2
6658
6788
6857
6884
8379000
8376000
8373000
8370000
8380000
8378000
8376000
8440000
8437500
8435000
8432500
8430000
8432000
8430000
8428000
8426000
0
20
40
60
80
value
day


MA(5) Non−Stationary Violations — Page 3
6921
7036
7043
7125
8448000
8447500
8447000
8446500
8446000
8445500
8440000
8438000
8436000
8434000
9300000
9100000
8900000
8400000
8360000
8320000
8280000
0
20
40
60
80
value
day


MA(5) Non−Stationary Violations — Page 4
7129
7395
7449
7477
value
day
8439000
8438500
8438000
8437500
8437000
8400000
8380000
8360000
8340000
8900000
8800000
8700000
8600000
8500000
8400000
8300000
8374000
8372000
8370000
0
20
40
60
80


MA(5) Non−Stationary Violations — Page 5
7487
7522
7533
7581
value
day
8260000
8250000
8240000
8230000
8220000
8600000
8550000
8500000
8450000
8535000
8530000
8525000
8420000
8400000
8380000
0
20
40
60
80



7.4.2 Double Moving Average: MA(5) → MA(10)

The remaining nonstationary series are smoothed again using an MA(10) applied to the MA(5) signal. This is intended to further isolate slower-moving calibration behavior from short-term fluctuations.

7.4.3 Test Stationarity After Double Smoothing

The MA(10)-on-MA(5) signal is tested using the ADF test.

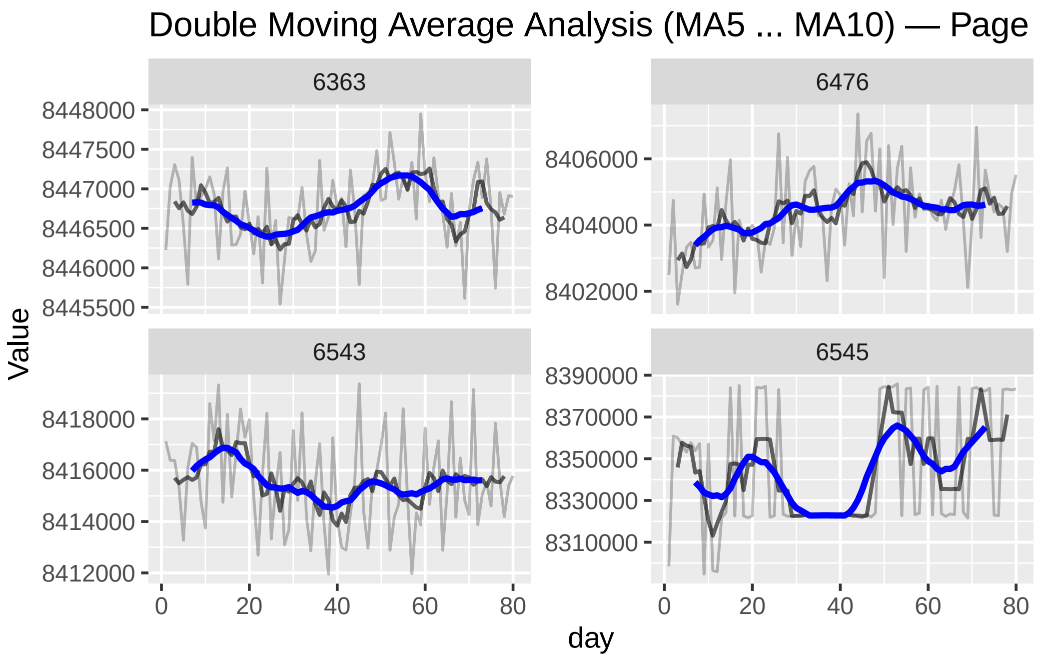
7.4.4 Summary of Double-Smoothing Results

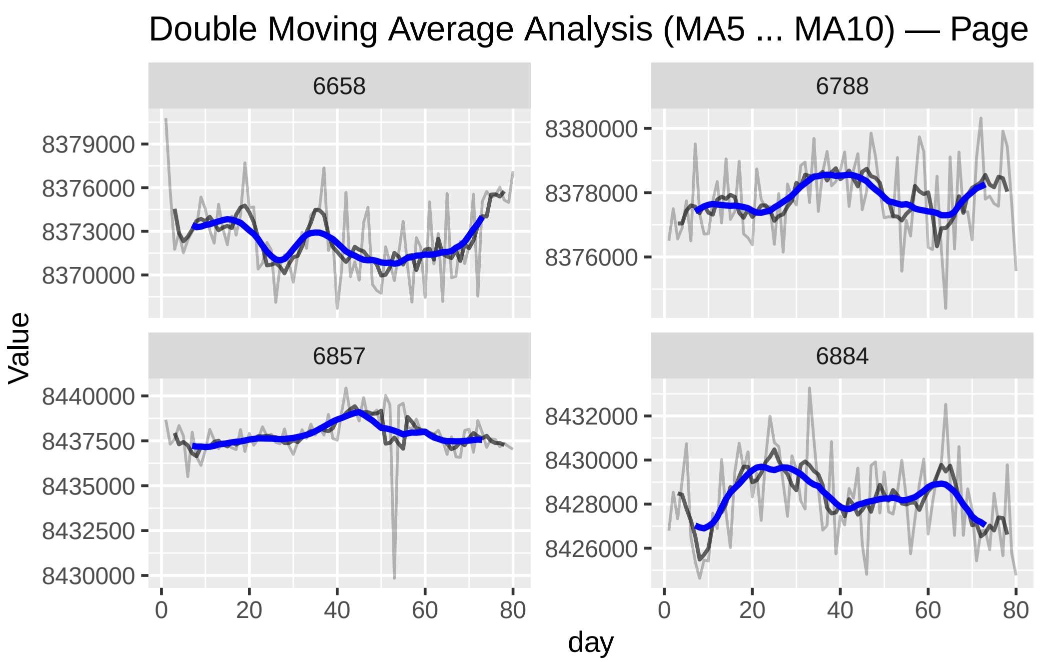
The proportion of the remaining series that become stationary after the second smoothing stage is calculated.

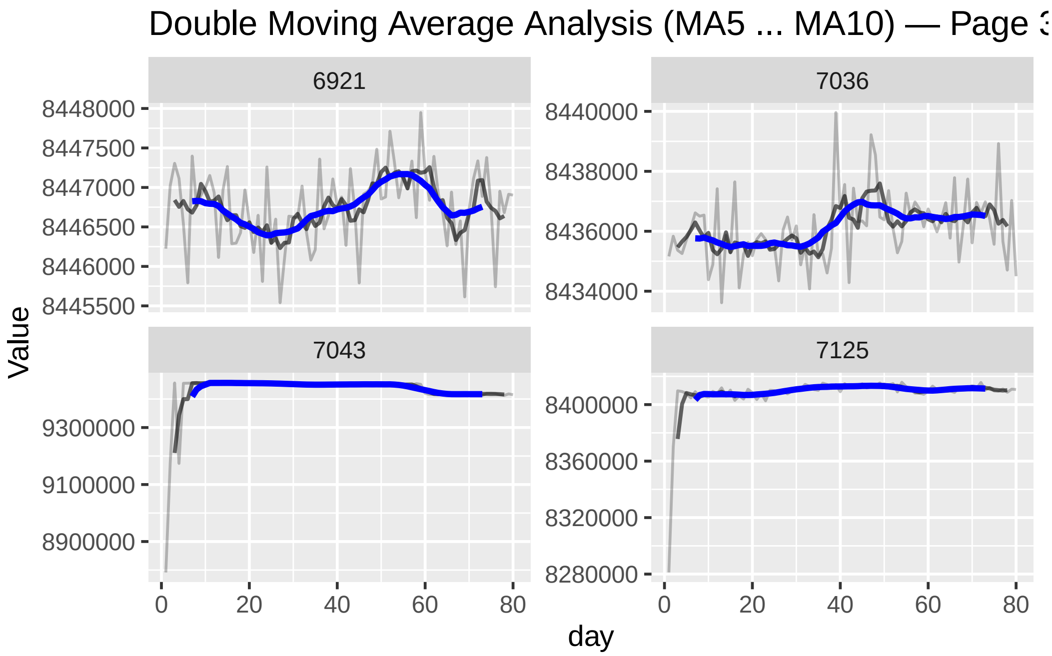
```
# A tibble: 1 x 3
  total stationary_after_double_ma proportion
  <int>                      <int>      <dbl>
1    24                          7      0.292
```

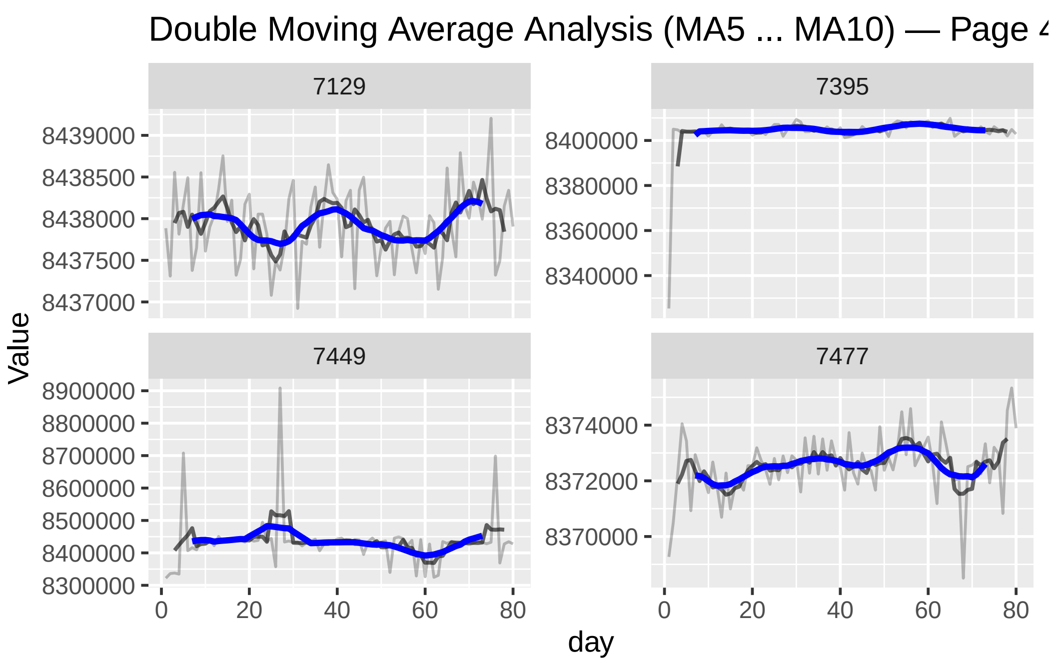
7.4.5 Visualize Double-Smoothing Results

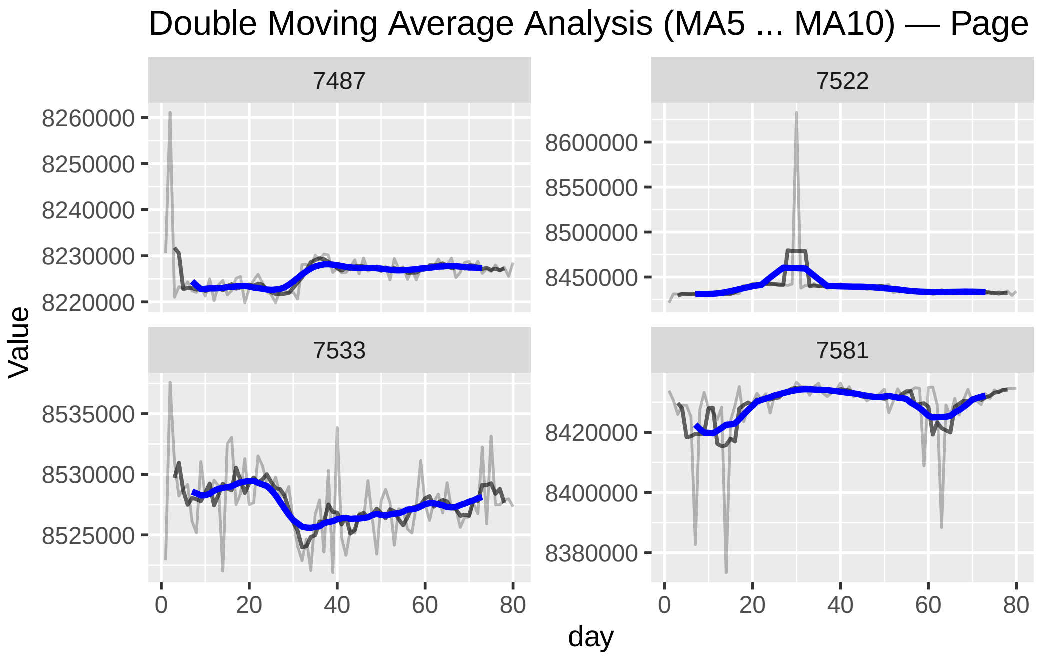
The original signal, MA(5), and MA(10)-on-MA(5) signals are plotted together.

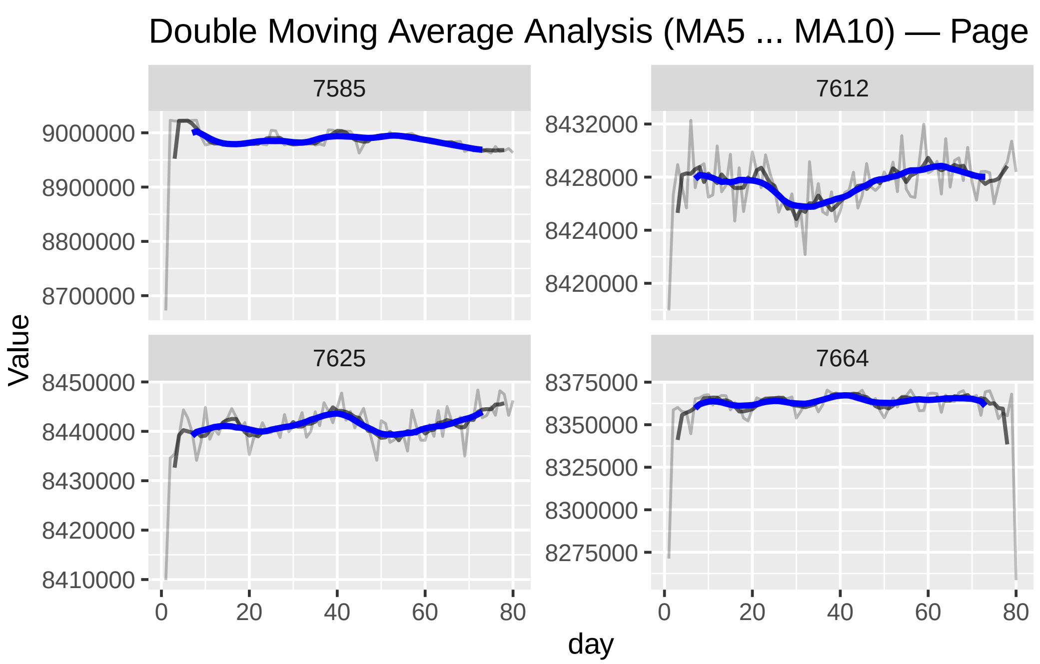





Double Moving Average Analysis (MA5 ... MA10) — Page
6921
7036
7043
7125
Value
day
8448000
8447500
8447000
8446500
8446000
8445500
8440000
8438000
8436000
8434000
9300000
9100000
8900000
8400000
8360000
8320000
8280000
0
20
40
60
80

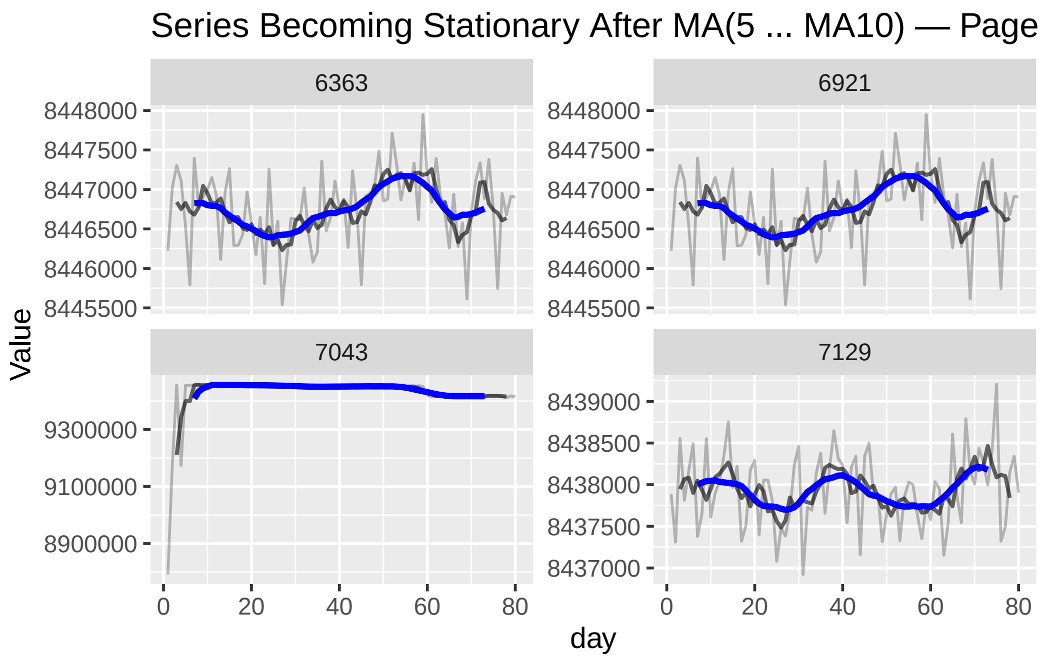

Double Moving Average Analysis (MA5 ... MA10) — Page
7129
7395
7449
7477
Value
day
8439000
8438500
8438000
8437500
8437000
8400000
8380000
8360000
8340000
8900000
8800000
8700000
8600000
8500000
8400000
8300000
8374000
8372000
8370000
0
20
40
60
80

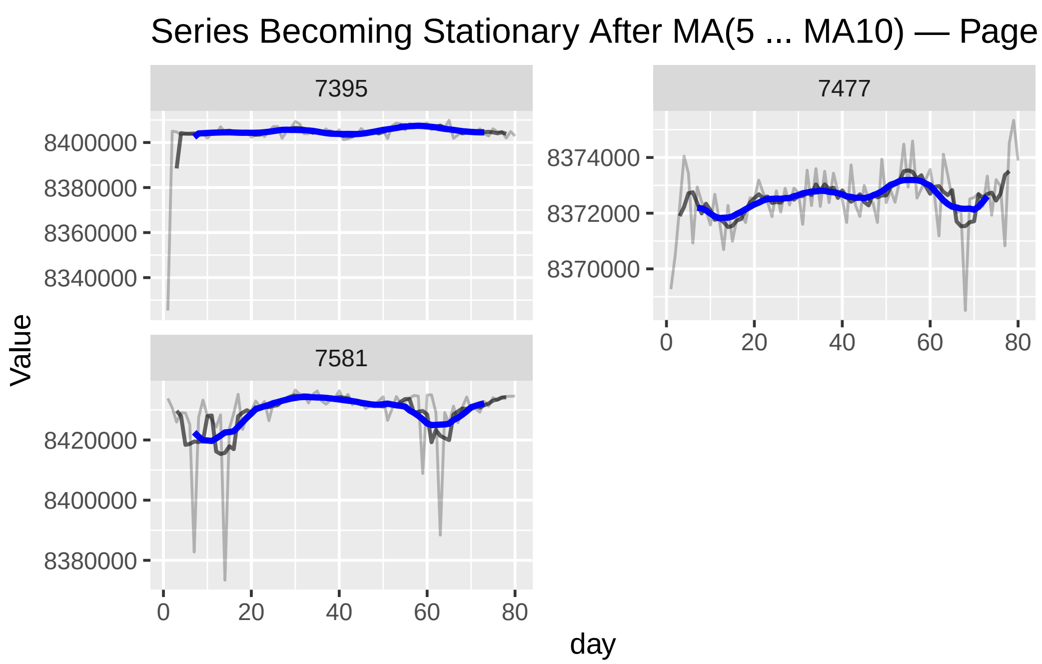

Double Moving Average Analysis (MA5 ... MA10) — Page 5
7487
7522
7533
7581
Value
day
8260000
8250000
8240000
8230000
8220000
8600000
8550000
8500000
8450000
8535000
8530000
8525000
8420000
8400000
8380000
0
20
40
60
80


Double Moving Average Analysis (MA5 ... MA10) — Page 6
7585
7612
7625
7664
Value
day
9000000
8900000
8800000
8700000
8432000
8428000
8424000
8420000
8450000
8440000
8430000
8420000
8410000
8375000
8350000
8325000
8300000
8275000
0
20
40
60
80

7.4.6 Series Becoming Stationary After Double Smoothing

The IDs that become stationary after the MA(5) → MA(10) process are isolated and visualized separately.





7.5 Third Smoothing Stage: Summary of MA(5) → MA(10) → MA(4) Results

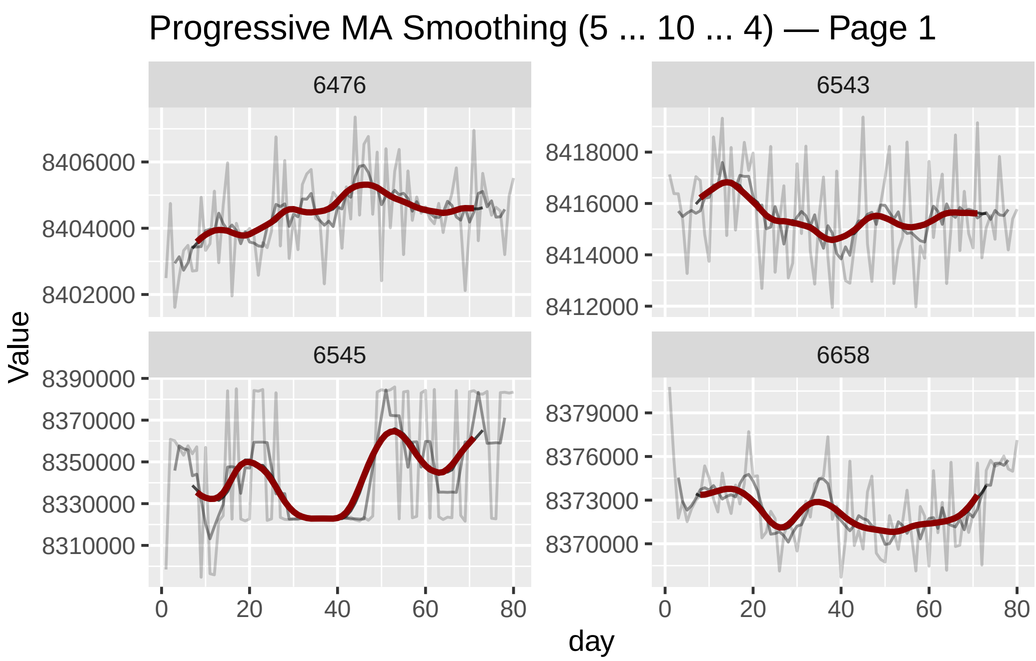
The series that remain nonstationary after the first two smoothing stages are isolated. A final MA(4) is applied to the already smoothed MA(10)-on-MA(5) signal.

7.5.1 Test Stationarity After Third Smoothing

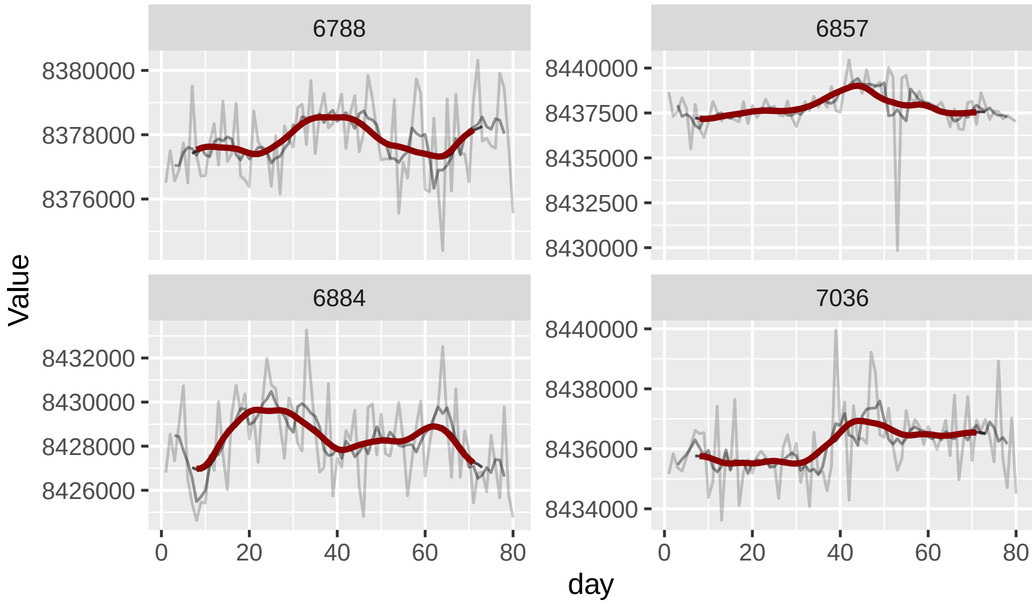
The final smoothed signal is tested for stationarity using the ADF test.

7.5.2 Visualize Progressive Smoothing

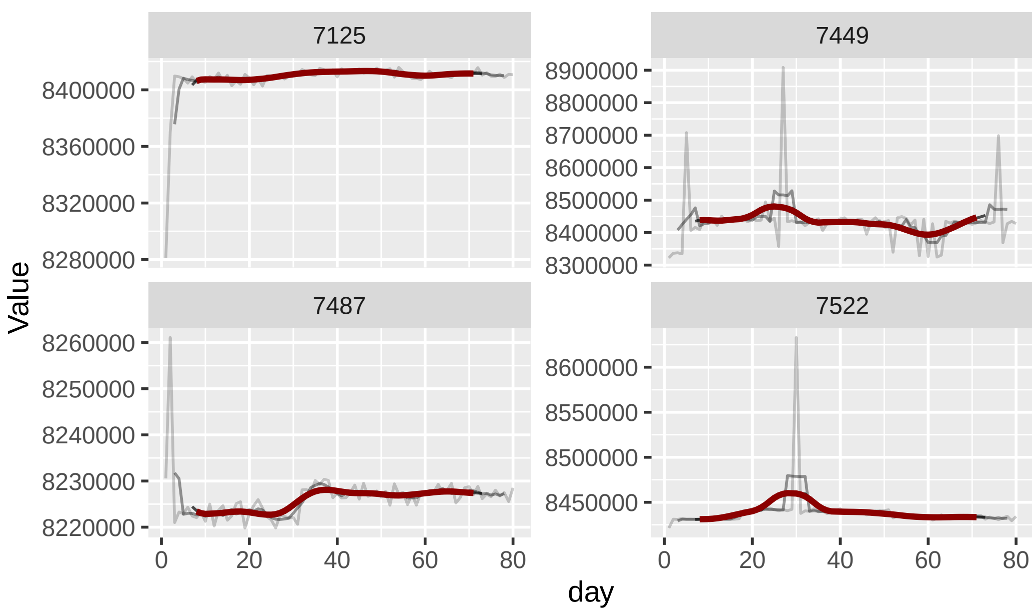
The original signal and all three smoothed versions are displayed together: original, MA(5), MA(10) applied to MA(5), and MA(4) applied to the previous smoothed signal.

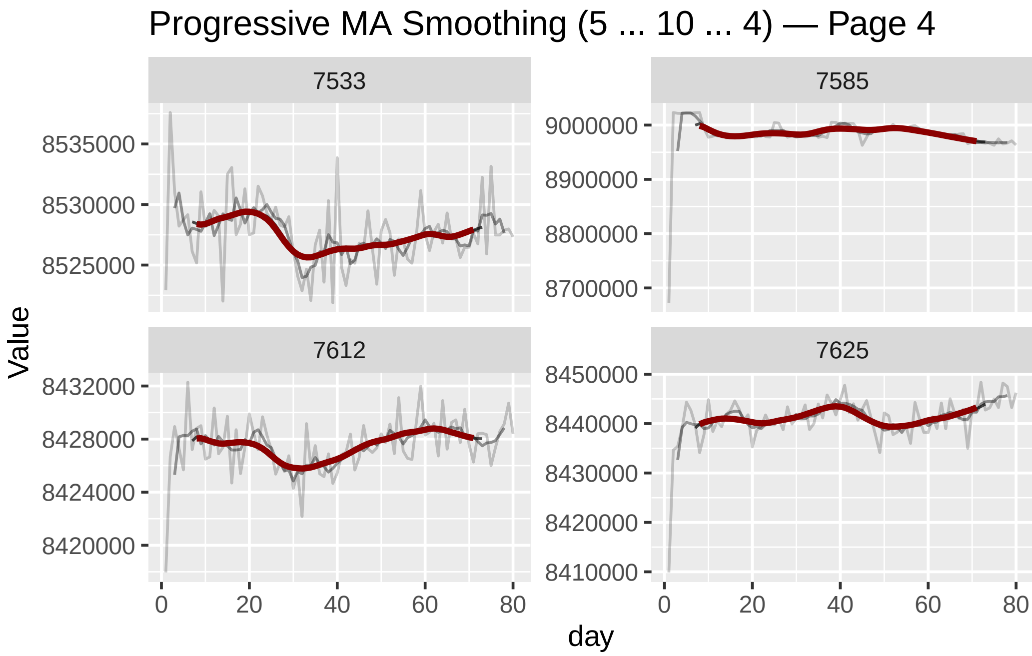


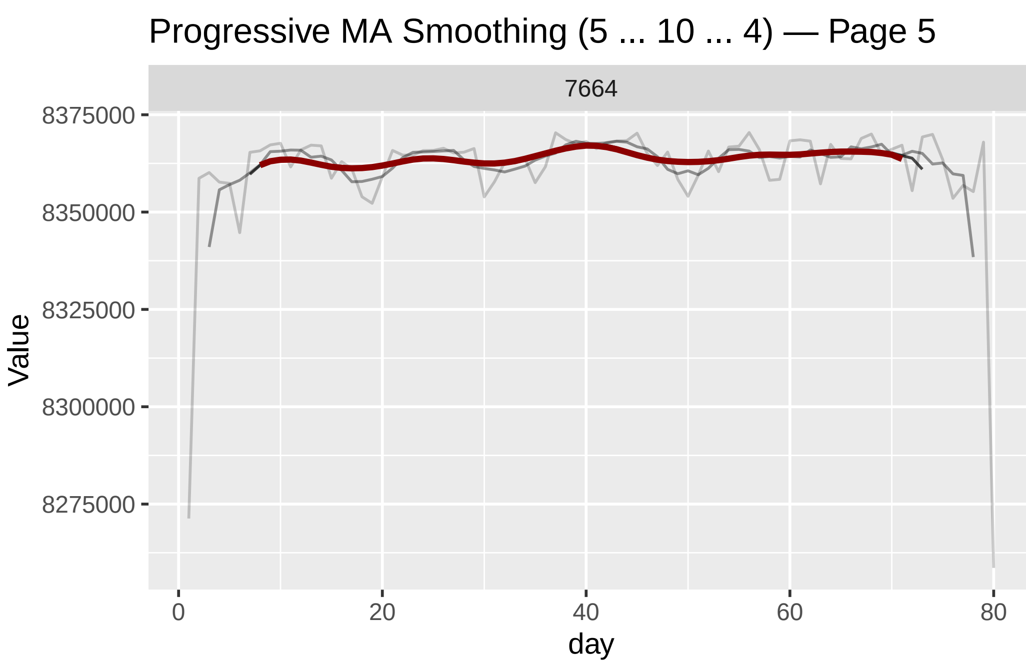
Progressive MA Smoothing (5 ... 10 ... 4) — Page 2



Progressive MA Smoothing (5 ... 10 ... 4) — Page 3




Progressive MA Smoothing (5 ... 10 ... 4) — Page 4
7533
7585
7612
7625
Value
day
8535000
8530000
8525000
9000000
8900000
8800000
8700000
8432000
8428000
8424000
8420000
8450000
8440000
8430000
8420000
8410000
0
20
40
60
80


Progressive MA Smoothing (5 ... 10 ... 4) — Page 5
7664
Value
day
8375000
8350000
8325000
8300000
8275000
0
20
40
60
80

7.5.3 Summary of Third Smoothing Stage

The number and proportion of remaining series that become stationary after the third smoothing stage are calculated.

```
# A tibble: 1 x 3
  total stationary_after_ma3 proportion
  <int>                <int>      <dbl>
1    17                    0          0
```

8. Interpretation: Moving Toward a Calibration Threshold

The progressive smoothing analysis is intended to determine whether the apparently non-stationary white-noise series contain an underlying, slowly changing calibration signal. If additional smoothing does not produce stationarity, the remaining variation may represent genuine long-term calibration behavior rather than short-term random noise.

The next step is therefore to examine the residual error around the final smoothing approach rather than continuing to smooth indefinitely.

8.1 Residual Analysis Around the MA(10) Signal

For the remaining series, the residual is defined as: residual = observed value - smoothed value. The residual standard deviation and variance quantify the amount of short-term variation around the underlying smoothed signal. The range and 80th percentile of the smoothed signal are also calculated as potential measures of the operating calibration range.

8.2 Global Residual Statistics

A population-level version of the residual statistics is also calculated. This provides an overall description of calibration variability rather than one statistic per ID.

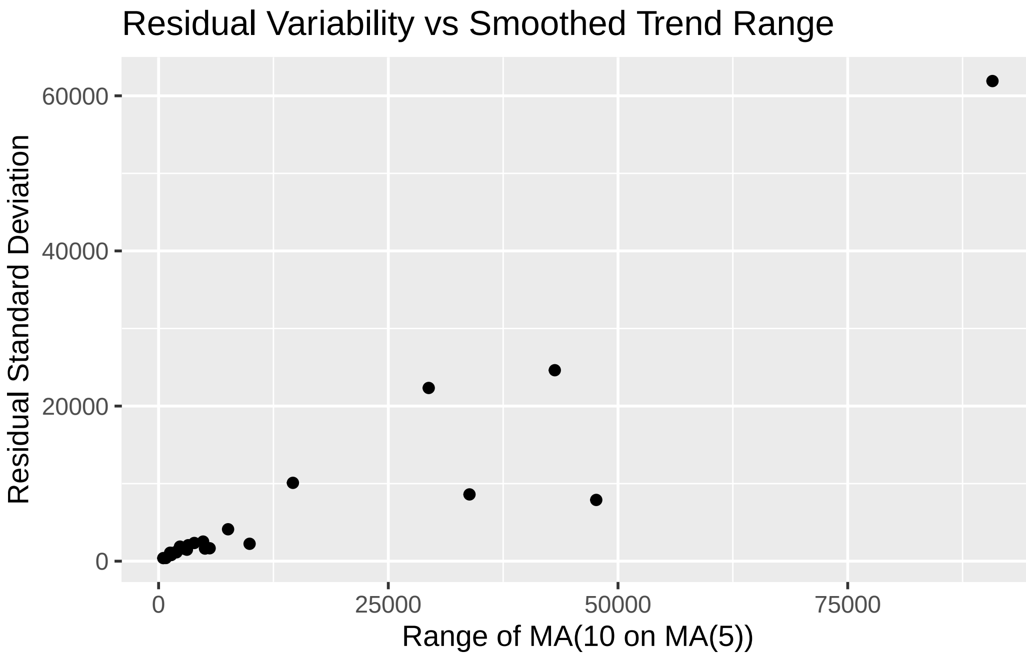
```
# A tibble: 1 x 4
  sd_residual_global var_residual_global range_ma2_global p80_ma2_global
               <dbl>               <dbl>            <dbl>          <dbl>
1             14668.          215160688.         1233953.       8446649.
```

8.3 Add Geographic Information to Residual Statistics

The residual statistics are joined to the landmark metadata so that residual behavior can be examined geographically if needed.

8.4 Residual Variability vs. Smoothed Calibration Range

The relationship between the range of the smoothed calibration signal and residual standard deviation is visualized. This helps determine whether series with larger underlying calibration movement also tend to have greater short-term variability.



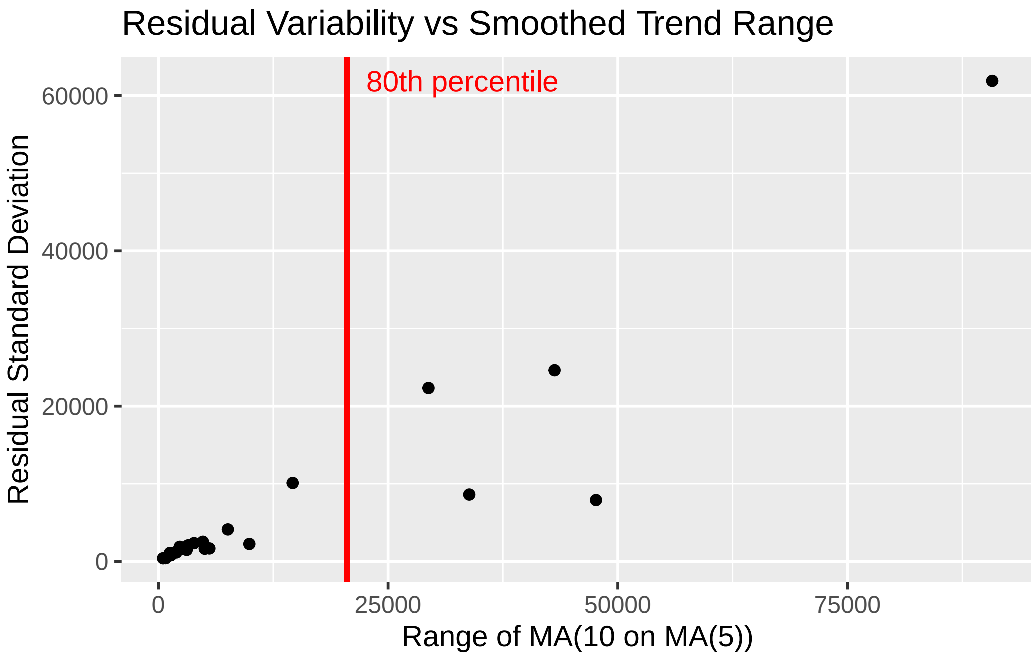
8.5 Conservative Range Threshold

The 80th percentile of the smoothed-value range is calculated. This is being explored as a potential conservative operating threshold for understanding how much the underlying calibration signal typically changes before a recalibration decision might be warranted.

```
     80%
20529.25
```

8.6 Visualize the 80th-Percentile Range Threshold

The 80th-percentile range is added to the residual-versus-range plot as a vertical reference line.


Residual Variability vs Smoothed Trend Range
80th percentile
60000
40000
20000
0
Residual Standard Deviation
0
25000
50000
75000
Range of MA(10 on MA(5))

9. Next Analytical Question

The analysis now provides the components needed to investigate a future recalibration rule:

the underlying calibration behavior estimated through smoothing, short-term residual variability around that behavior, the amount of normal calibration movement, the distribution of calibration ranges, the stationary/nonstationary classification, and detected change points for unstable series.

A proposed next step is to determine when the observed calibration behavior has moved far enough from its expected operating behavior to justify recalibration.

For the nonstationary white-noise series, the current analysis suggests using the smoothed signal as an estimate of underlying behavior and the residual distribution as an estimate of normal short-term calibration variation.

A conservative percentile such as the 75th or 80th percentile could then be evaluated as a candidate threshold. The appropriate percentile and the exact recalibration trigger still require validation against the actual calibration performance and desired false-alarm/recalibration tradeoff.

For non-white-noise, nonstationary series, the next step is to investigate whether smoothing can reveal piecewise-stationary periods. Those periods can then be evaluated to determine when the calibration behavior changes enough to warrant recalibration.